

August 20, 2026 09:03 PM GMT

Global Credit Strategy & Securitized Products Research

AI Debt Financing Tracker: The Summer Spreads Turned Wider

HQ Hyperscalers are ~30bp wider YTD, yet supply has not slowed, with \$54bn done in July and \$41bn in August (MTD). Even with ~\$445bn of issuance across credit channels YTD, we expect a busy fall and continued spread widening. In addition to public market supply, HQ issuers will continue to flex their ratings muscles to provide credit support to others as new financing structures emerge. Collateral-backed & stabilized assets have better risk-reward than unsecured corporates.

Key Takeaways

- We saw ~\$92bn of AI-related new issuance in corporate credit globally since July, which reinforces our view that market capacity remains ample, but price sensitivity and the consequences to issuance vary across the credit spectrum. Wider spreads – especially for HQ borrowers – are the natural release valve for corporate credit markets to continue to fund the AI capex cycle.
- Over \$20bn of debt backed by data centers came to market in July alone, pushing spreads in corporate credit – but not securitized credit – meaningfully wider as the market continued to reprice elevated supply expectations. A steady stream of headlines on data center development pushback and policy measures that could slow down the pace of new assets from Texas to Ohio added to the noise on bonds with higher construction risk.
- In corporate credit, we think issuance out of hyperscalers could slow in the near term, but we would fade a move back to the tights given 2027 capex estimates (~\$14tr). We also expect high-quality names to leverage their AAA/AA ratings and provide credit support to the AI-ecosystem and facilitate access to capital to lower-quality/unrated/private companies that are more sensitive to higher costs of capital.
- In securitized credit, spreads held flat, largely shrugging off the selloff in credit. Issuance continues at a steady pace, though less strong than credit at ~65% of FY25 volumes. The stabilized nature of the assets backing ABS/CMBS insulates them from some of the risks causing volatility in corporate credit. That said, we think risks skew to the downside from here and expect to see spreads soften in the fall once supply resumes.
- While elevated supply and wider spreads have been largely top of mind, clients seem increasingly focused on the financing structures themselves particularly given the recent NVDA financing platform announcement. We expect HQ credit support to play a bigger role in private capital engagement in the ongoing AI infrastructure buildout, especially as capex shifts from powered shells to components like chips/servers.

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We're changing our publication format and expanding our content. For detailed issuance breakdown, refer to [Exhibit 3](#) ; for details on deals priced since mid-July, see [Exhibit 4](#) . We also expanded our [relative value analysis](#) for data center bonds given the overlap between corporate and securitized credit. [Capex estimates](#) now include SPCX.

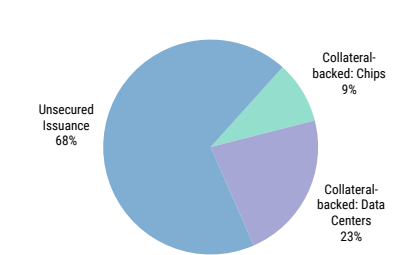
- For **US Investment Grade**, see the [HQ hyperscaler curves vs. AA/A/BBB curves](#), as well as a [tracker for largest unsecured corporate issuers](#) and more detail on [CDS performance](#).
- For **Investment Grade Reverse Yankees**, we continue to [track performance of non-USD deals](#), and we're now looking at [maturity profile](#) of issuance as well.
- Within **Leveraged Finance**, we're expanding our relative value work on [HY data center bonds](#) and tracking the [pickup in loan issuance](#).
- As for **Securitized Credit**, we discuss the recent regulatory changes, introduce additional content on [spread performance for ABS](#) and [CMBS](#) and incorporate specific content on non-USD securitized credit.
- For **Single Name**, ORCL and AVGO remain front and center ahead of their respective September earnings prints.

Exhibit 1: Summary of YTD issuance and estimates

Issuance (USD bn)	YTD-26	FY26E	FY25A
HQ Hyperscalers	132		67
Oracle	25		26
Hyperscaler Total	157	250-300	93
Other AI Total	114	100	64
Public IG Total	271	350-400	157
4a2 Priv. Plac.	35	-	0
Total HY	40	50	15
Secd Lev. Loans	10	15	2
Total US Corporate Credit	356	415-465	174
Total US Securitized Credit	18	33	27
Corporate Credit non-USD	69	50	15
Securitized Credit non-USD	2	2	1
Total non-USD	70	52	16
Total Global Credit Markets	445	500-550	217

Source: Bloomberg, Pitchbook | LCD, CreditFlow, TREPP, Morgan Stanley Research estimates. Note: YTD 2026 data includes deals priced until 8/19/26. For a more detailed version of this exhibit, refer to [Exhibit 2](#) . Figures are rounded to the nearest \$bn; totals are calculated using unrounded figures and may differ slightly from the sum of displayed values.

Exhibit 2: Majority of AI-related debt is unsecured corporate issuance



Source: Bloomberg, Company filings, Morgan Stanley Research. For data center breakdown and rel val, see [Exhibit 9](#) and [Exhibit 5](#) .

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- Other relevant content on AI financing:
- [US Credit Strategy: A Break from the Heat, For Now \(14 Aug 2026\)](#)
 - [Data Center ABS: Finding Value Amidst A Credit Market Sell-off \(6 Aug 2026\)](#)
 - [Sunday Start | What's Next in Global Macro: Summer of AI Financing \(16 Aug 2026\)](#)
 - [Power Struggle: Data Center Pushback Update & Expert Call Takeaways \(17 Aug 2026\)](#)
 - [Power Struggle: AI, Data Centers & Local Backlash \(14 Jul 2026\)](#)
 - [TMT Credit Research: A Rising Backstop Doesn't Lift All Credits \(28 Jul 2026\)](#)

Issuance Summary & Estimates

Exhibit 3: We're now at \$445bn of AI-related debt issuance across global credit markets; despite spread widening, July and August were busy months in terms of issuance

Markets			Issuers <i>USD bn, unless otherwise noted</i>	Year-to-Date Issuance Tracker									
				Jan-26	Feb-26	Mar-26	Apr-26	May-26	Jun-26	Jul-26	Aug-26 (MTD)	2026 YTD	2025 YTD
Corporate Credit (U.S.)	U.S. Investment Grade	HQ Hyperscalers	0	20	37	25	0	0	25	25	132	5	2540%
		Oracle	0	25	0	0	0	0	0	0	25	8	223%
		SpaceX	0	0	0	0	0	25	0	0	25	0	-
		Semiconductors	5	0	0	8	0	25	0	5	42	27	57%
		REITS	0	2	0	0	0	0	3	0	5	0	-
		Secd Data Center Const.	0	0	0	22	0	4	13	4	43	0	-
		Total IG	5	47	37	54	0	54	41	34	271	39	588%
	U.S. Leverage Finance	HY Secd Data Center Const.	0	6	2	16	0	4	4	2	34	0	-
		HY Corporate	0	0	0	3	0	3	1	0	7	7	-4%
		Secd Loans	0	0	0	0	3	0	6	1	10	2	394%
Total LevFin		0	6	2	19	3	7	10	3	50	9	471%	
U.S. 4a2 Private Placement			0	0	0	0	0	35	0	0	35	0	-
Total U.S. Corporate Credit			5	52	39	73	3	96	51	37	356	48	640%
Securitized Credit (U.S.)	Securitized Credit	ABS	2	3	3	1	0	2	1	0	12	8	49%
		CMBS	0	2	0	0	1	2	1	0	6	7	-14%
	Total U.S. Securitized Credit		2	5	3	1	1	4	2	0	18	15	20%
Total U.S. Credit Markets			7	57	42	74	4	100	53	37	374	63	490%
Corporate Credit (non- USD)	IG Hyperscalers	EUR ¹	0	0	17	0	11	0	0	0	27	8	255%
		GBP ¹	0	8	0	0	0	0	0	0	8	0	-
		CAD ¹	0	0	0	0	6	10	0	0	16	0	-
		CHF ¹	0	4	0	0	4	0	0	0	8	0	-
		JPY ¹	0	0	0	0	4	0	0	0	4	0	-
		AUD ¹	0	0	0	0	0	0	0	4	4	0	-
	Total IG Other Currencies		0	11	17	0	24	10	0	4	66	8	761%
Total HY Other Currencies		0	0	0	0	0	2	0	0	2	0	-	
Securitized Credit (non- USD)	Securitized Credit	ABS	0	1	0	0	0	0	0	0	1	1	-1%
		CMBS	0	0	1	0	0	0	0	0	1	0	-
	Total Non-USD Securitized Credit		0	1	1	0	0	0	0	0	2	1	165%
Total Global Credit Markets			7	69	59	74	28	112	54	41	445	72	519%

Notes: Includes transactions priced until 8/19/26 EoD. YTD 2025 figures include August MTD; except for US SPG that we're including data until 7/31/25. Presented figures exclude IG TLs such as AMZN DDTL. 1. All figures in USD-equivalent billions.

Source: Bloomberg, PitchBook | LCD, CreditFlow, TREPP, Morgan Stanley Research estimates. Note: YTD 2026 data includes deals priced until 8/19/26. Figures are rounded to the nearest \$bn; totals are calculated using unrounded figures and may differ slightly from the sum of displayed values.

Exhibit 4: Outside of HY, deals priced since mid-July have generally tightened since issuance

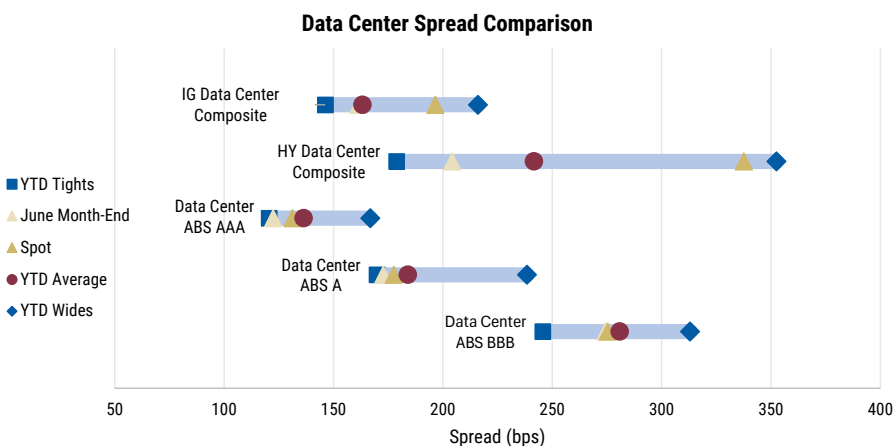
Ticker	Issue Date	Maturity Date	Size (\$bn)	Composite Rating	Coupon	Issue Spread to Benchmark (bps)	Current Spread to Benchmark (bps)	Spread Δ Since Issuance (bps)
Corporate Credit - Investment Grade								
TXEDCI	7/27/2026	11/30/2048	12.6	A+	7.53%	288	251	-37
EQIX	7/30/2026	8/15/2031	0.9	BBB+	5.25%	95	96	+1
EQIX	7/30/2026	8/15/2036	0.7	BBB+	5.80%	120	127	+7
GOOGL	8/6/2026	8/15/2031	3.5	AA	4.88%	60	56	-5
GOOGL	8/6/2026	8/15/2036	4.5	AA	5.45%	85	81	-5
GOOGL	8/6/2026	8/15/2056	4.5	AA	6.38%	115	110	-5
AMD	8/13/2026	8/17/2031	1.5	A	5.00%	70	69	-1
AMD	8/13/2026	8/17/2036	1.0	A	5.50%	90	100	+10
QTSODY	8/18/2026	9/1/2031	3.9	N.R. ¹	6.63%	286	262	-25
Corporate Credit - High Yield								
GALAXY	7/23/2026	8/1/2031	3.5	BB-	9.88%	553	560	+7
ZENARC	8/13/2026	8/31/2031	2.3	N.R. ²	8.88%	467	560	+93
Corporate Credit - Leverage Loans								
QTS (S+225)	7/15/2026	7/25/2033	3.3	BBB-	5.90%	236	231	-5
CRWV (S+550)	7/30/2026	9/30/2031	2.6	BB	9.15%	584	527	-57
LAMBCM	8/12/2026	12/31/2030	0.9	N.R. ³	6.65%	303	287	-16
Corporate Credit - Investment Grade Reverse Yankee⁴								
GOOGL	8/19/2026	8/27/2031	0.5	N.R.	5.50%	90	-	-
GOOGL	8/19/2026	8/27/2036	0.7	N.R.	6.25%	133	-	-
GOOGL	8/19/2026	8/27/2046	0.7	N.R.	6.90%	180	-	-
Securitized Credit - ABS⁵								
CMDC 2026-2	7/24/2026	7/25/2031	0.3	AAA	5.72%	135	133	-2
CMDC 2026-2	7/24/2026	7/25/2031	0.1	AA-	6.06%	170	165	-5
CMDC 2026-2	7/24/2026	7/25/2031	0.1	A	6.30%	195	184	-11

Notes: 1. QTSODY ratings are Baa3 by Moody's and BBB- (Exp) by Fitch. ZENARC was rated Ba2 by Moody's and BB (Prelim) by S&P. 3. LAMBCM TL was rated Baa1 by Moody's. 4. Issue spread is based on spread to swap (sqASW + reflected spread). 5. For ABS, maturity date reflects the anticipated repayment date (ARD).

Source: Bloomberg, CreditFlow, Green Street, Morgan Stanley Research. Current levels as of 8/19/26.

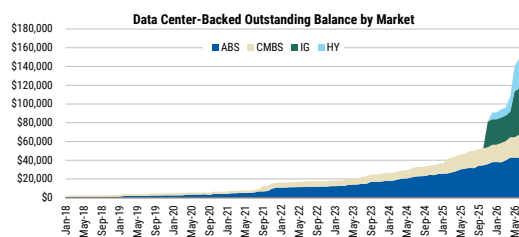
Data Center Relative Value: Corporate Credit vs. Securitized Credit

Exhibit 5: Data center-backed paper in corporate credit widened over the last six weeks while ABS paper remained relatively steady



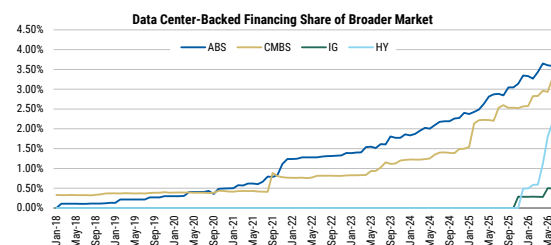
Source: Bloomberg, Morgan Stanley Research

Exhibit 6: Issuance of DC-backed paper has ramped quickly across markets this year, with the biggest gains in IG and HY



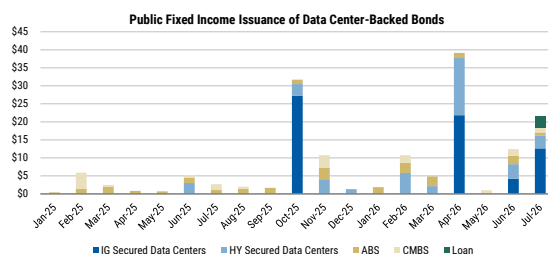
Source: Bloomberg, CreditFlow, Trepp, Intex, Morgan Stanley Research

Exhibit 7: Despite heavy supply all year, as a share of the broader markets DC-backed paper remains a very small proportion



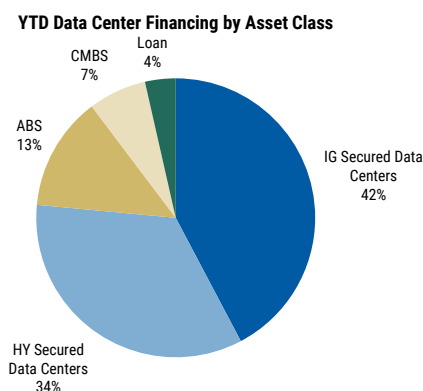
Source: Bloomberg, CreditFlow, Trepp, Intex, Morgan Stanley Research

Exhibit 8: Monthly issuance of data center-backed paper by product



Source: Bloomberg, CreditFlow, Trepp, Morgan Stanley Research

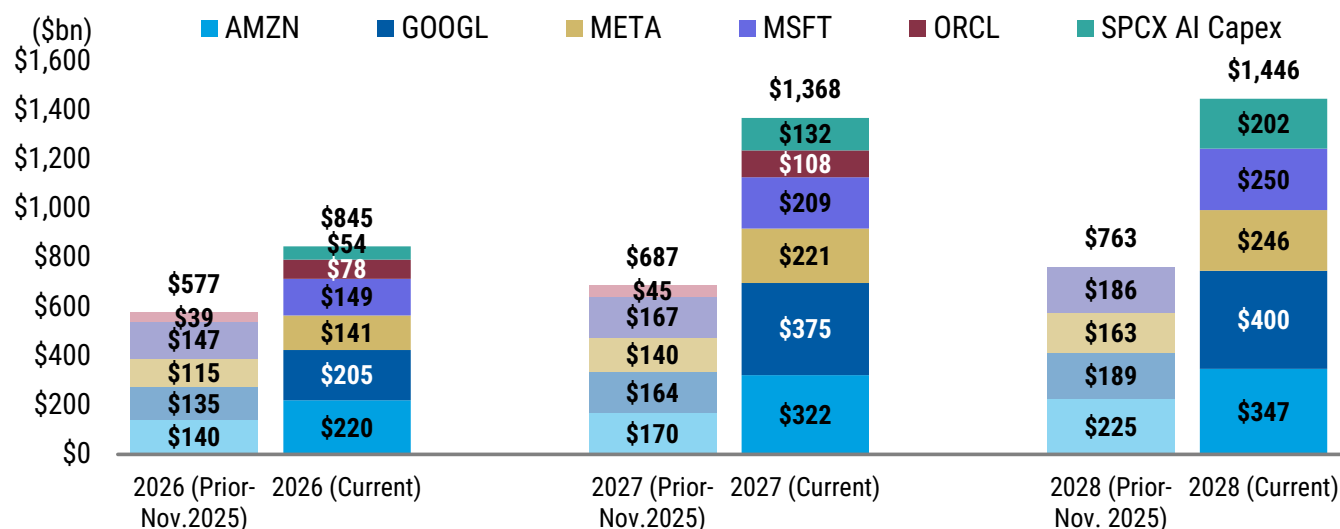
Exhibit 9: YTD financing breakdown of data center-backed paper



Source: Bloomberg, CreditFlow, Trepp, Morgan Stanley Research

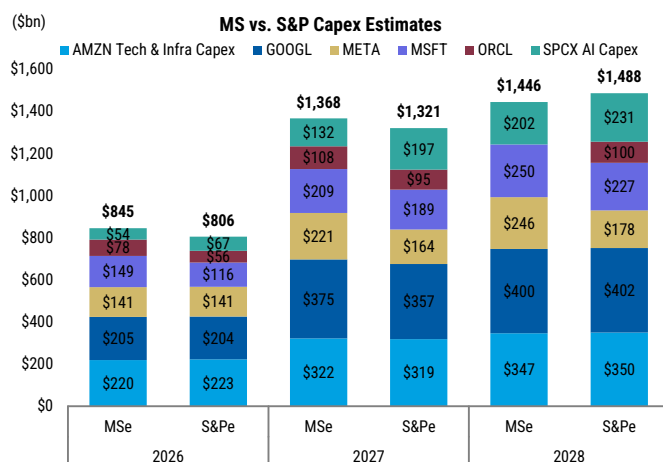
2027 Combined Capex Estimates Are Now Close to \$1.4tr

Exhibit 10: We're now including SPCX AI-related capex into our estimates; even when looking at just the 5 hyperscalers, 2027 combined capex estimates have nearly doubled Nov. 2025 estimates



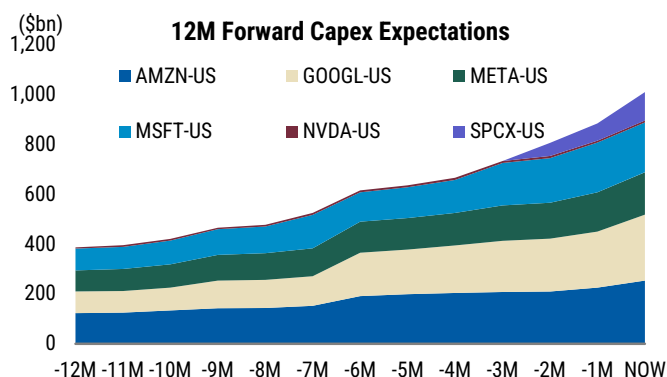
Source: Morgan Stanley Research current estimates from models dated as of: 7/31/26 for AMZN, 7/23/26 for GOOGL, and 7/30/26 for META (covered by Brian Nowak); 8/9/26 for MSFT, 4/23/26 for ORCL, and 8/11/26 for SPCX (covered by Adam Wood, Sanjit Singh, and Adam Jonas respectively). Prior estimates are from November 2025. Note: Includes current capital lease payments for MSFT and META. GOOGL/ORCL/AMZN are only cash capex. Company-level capex figures are rounded to the nearest \$bn; total capex is calculated using unrounded figures and may differ slightly from the sum of displayed labels.

Exhibit 11: After 2Q earnings, S&P's estimates are more in line with MSe, with some 2028 estimates higher



Source: S&P, Morgan Stanley Research estimates

Exhibit 12: NTM capex expectations have nearly tripled from a year ago



Source: FactSet, Morgan Stanley Research

US Investment Grade Credit: Wider Spreads Are the Release Valve to Accommodate Supply

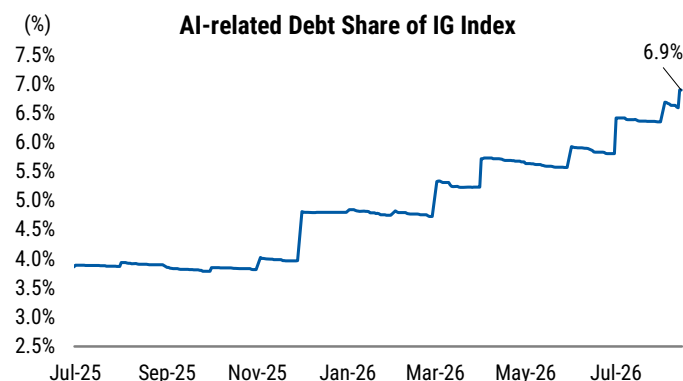
Within IG, two topics dominated our conversations during the past month (see below).

Refer to our recent note for our detailed views: [US Credit Strategy: A Break from the Heat, For Now \(14 Aug 2026\)](#).

- **Market capacity:** US IG is one of the deepest capital markets globally, with ~US \$10tn in size, predominantly owned by long-term institutional investors that have a structural need for high-quality, income-generating assets. With AI-related debt at ~7% of the IG index, we think there's still significant capacity to absorb high-quality paper at the right price, and think of ~15% of the index as a maximum sector exposure that investors would be comfortable with. We expect the public IG market to continue to absorb the bulk of AI issuance, in line with YTD trends (~61% of issuance was in the public US IG market).
- **Balance sheet capacity:** HQ hyperscalers have ample debt capacity as they're entering this cycle with low leverage, ample liquidity, double-digit growth rates on EBITDA, and cash from ops. Off-balance sheet commitments will impact leverage ([Exhibit 23](#)), and disclosures are limited, but they're included by rating agencies in their adjusted debt calculations. To the extent those increase, there is less issuance capacity at current rating levels. We also don't think HQ hyperscalers are sensitive to the move we saw in spreads given their expected returns; however, lower-quality names could be more impacted.

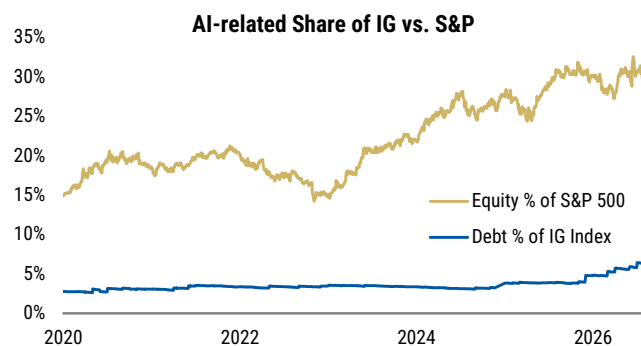
AI-Related Issuance Is Now ~7% of the IG Index; We Still See Ample Market Capacity from Current Levels

Exhibit 13: AI-related debt share of IG Index continues to rise rapidly on the back of heavy issuance



Source: Bloomberg, ICE BAML, Morgan Stanley Research. Note: AI-related includes a broader universe than just hyperscalers.

Exhibit 14: The 10 companies account for 7% of the IG public index, vs. ~30% of the S&P 500



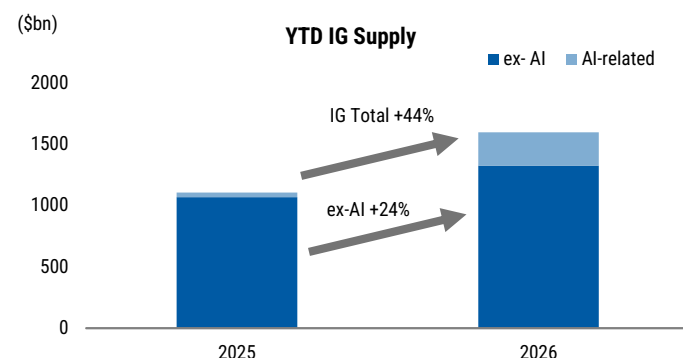
Source: Bloomberg, ICE BAML, Morgan Stanley Research

Exhibit 15: Hyperscalers now have \$431bn of index-eligible debt; other large AI names add ~\$110bn

Issuer	Index Eligible Debt Out (\$bn)	% of IG Index	IG Index Rank	Composite Rating
AMZN	121	1.2%	5	AA3
ORCL	117	1.2%	6	BBB2
META	84	0.8%	11	AA3
GOOGL	76	0.8%	13	AA2
MSFT	34	0.3%	54	AAA
Hyperscaler Total	431	4.3%		
AVGO	54	0.5%	21	A3
NVDA	33	0.3%	61	AA2
SPCX	25	0.2%	95	BBB1
Total	543	5.4%		

Source: ICE, Morgan Stanley Research; Note: Sorted by Index debt outstanding.

Exhibit 17: The uptick in IG Supply YTD is largely driven by AI-related financing



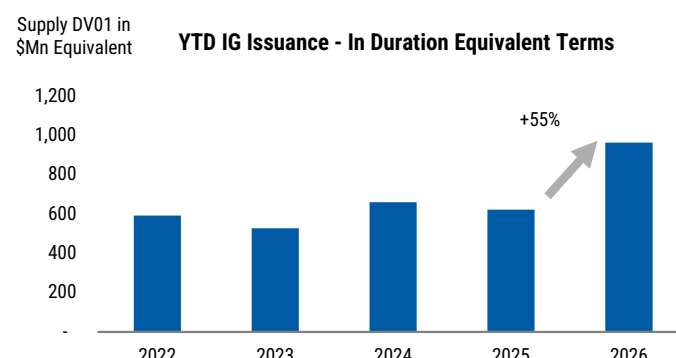
Source: Dealogic, Morgan Stanley Research

Exhibit 16: GOOGL joins top 15 largest issuers with August issuance at #13

Rank	Ticker	Sector	Index Debt	% of Index
1	JPM	Banking	179	1.8%
2	MS	Financial Services	169	1.7%
3	BAC	Banking	166	1.7%
4	GS	Financial Services	141	1.4%
5	AMZN	Retail	121	1.2%
6	ORCL	Technology & Electronics	117	1.2%
7	WFC	Banking	106	1.1%
8	C	Banking	105	1.1%
9	T	Telecommunications	101	1.0%
10	HSBC	Banking	89	0.9%
11	META	Media	84	0.8%
12	VZ	Telecommunications	82	0.8%
13	GOOGL	Media	76	0.8%
14	CMCSA	Media	74	0.7%
15	UNH	Healthcare	70	0.7%

Source: ICE, Morgan Stanley Research; Note: Sorted by Index debt outstanding. Index eligible debt rebalanced monthly and may not include intramonth issuance (GOOGL's August issuance is included).

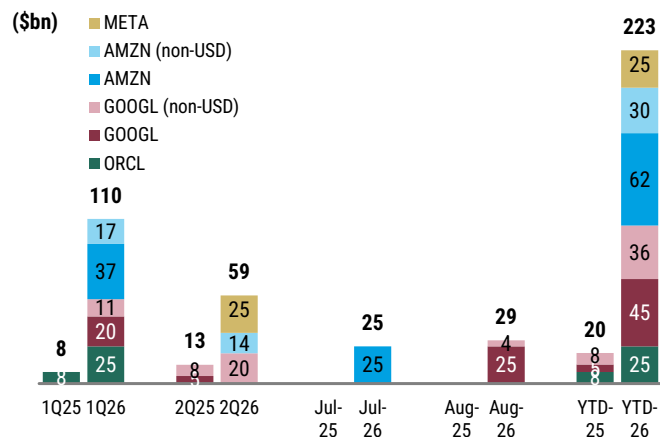
Exhibit 18: In duration terms, overall IG issuance is up 55% y/y due to AI deals



Source: Dealogic, Morgan Stanley Research; Note: Issuance converted to duration terms in Mn (Notional *duration/100)

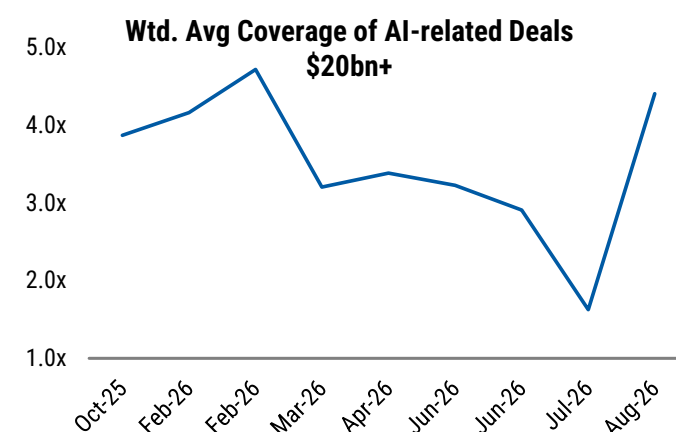
Hyperscalers Funding Mix & Balance Sheet Capacity

Exhibit 19: YTD issuance for the hyperscalers is at \$223bn, with ~\$66bn of non-USD debt



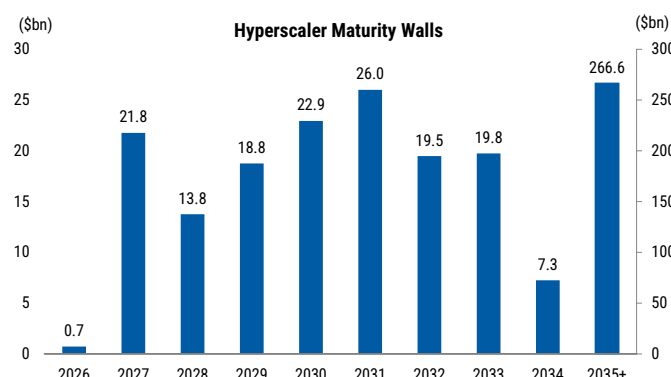
Source: Bloomberg, Morgan Stanley Research

Exhibit 20: While average coverage across the 9 large AI-related deals we have seen since 4Q25 is 3.5x, timing matters for individual deals



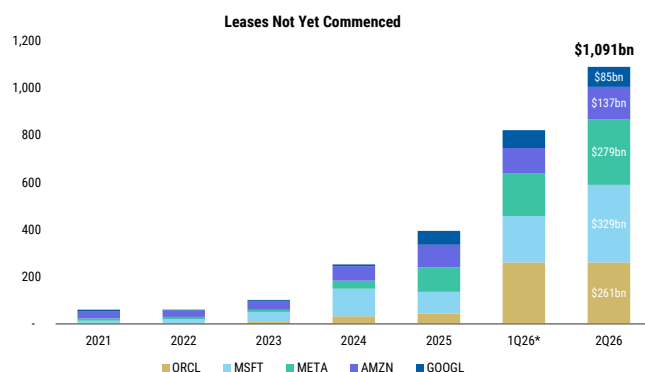
Source: Bloomberg, Morgan Stanley Research

Exhibit 21: Hyperscalers have ~\$22.5bn in bond maturities between now and YE27



Source: ICE BAML, Morgan Stanley Research

Exhibit 23: Off-balance sheet leverage is rising as hyperscalers enter into long-dated leases for data center shells...



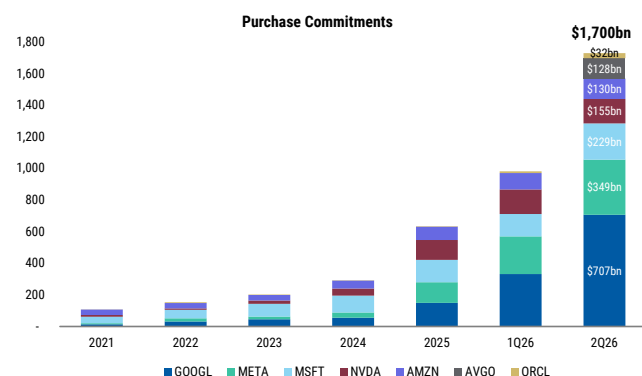
Source: Company Filings, Morgan Stanley Research.

Exhibit 22: Hyperscaler gross leverage is now 1.3x, however net leverage still well below 1x

Select Metrics by Sector 2Q26				
Sector	Gross Leverage	Net Leverage	Cash/Debt	Median Rating
Communication Services	2.0x	1.2x	49%	BBB+
Consumer	2.5x	1.8x	18%	BBB
Energy	1.6x	1.0x	19%	BBB
Health Care	2.7x	2.2x	17%	BBB+
Information Technology	1.6x	0.8x	56%	BBB
Hyperscalers	1.3x	0.4x	132%	AA-
Materials	2.2x	1.9x	14%	BBB
Utilities	5.8x	5.6x	1%	BBB
Non-Fin IG	2.4x	1.8x	16%	BBB

Source: FactSet, Bloomberg, Morgan Stanley Research. Note: Medians and averages for companies that reported 2Q26 earnings as of 8/7/26, roughly 80% of total universe. Note: Leverage numbers based on reported debt and does not include off-balance sheet commitments. For off-balance sheet commitments please see [Exhibit 23](#)

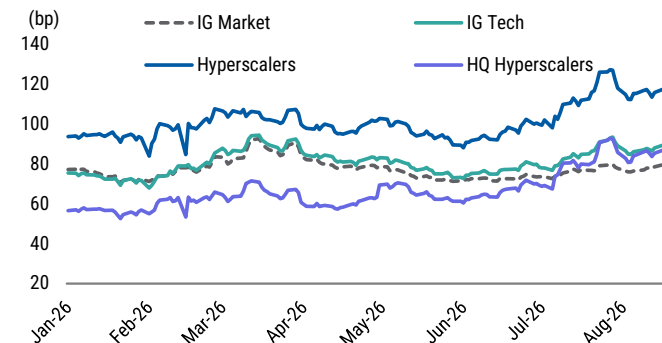
Exhibit 24: ...and commit to larger and longer-term purchases of compute, chips, equipment, and power



Source: Company Filings, Morgan Stanley Research.

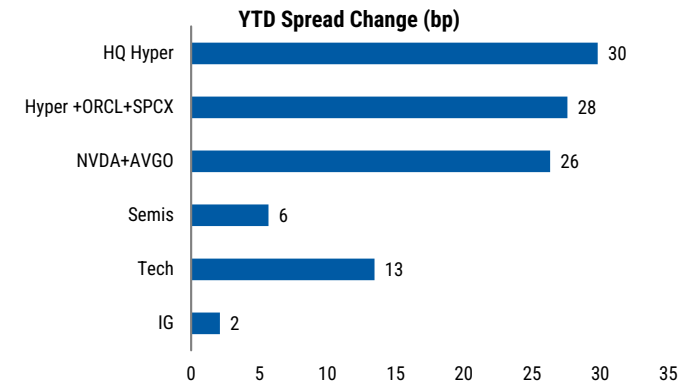
Hyperscaler Spread Performance: Hyperscalers Already Trade Wider to Their Current Ratings; We Think Volatility Is Likely to Continue

Exhibit 25: HQ Hyperscalers spreads are now 7bp wider than the IG Index



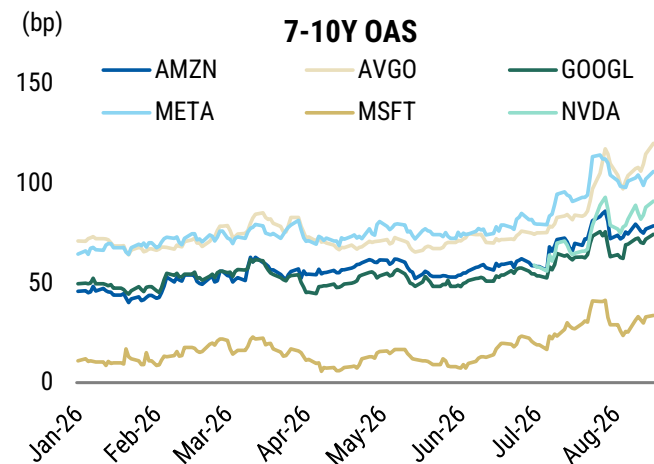
Source: Bloomberg, Morgan Stanley Research.

Exhibit 26: YTD widening most severe for the HQ Hyperscaler cohort, IG Index still only 2bps wider on the year



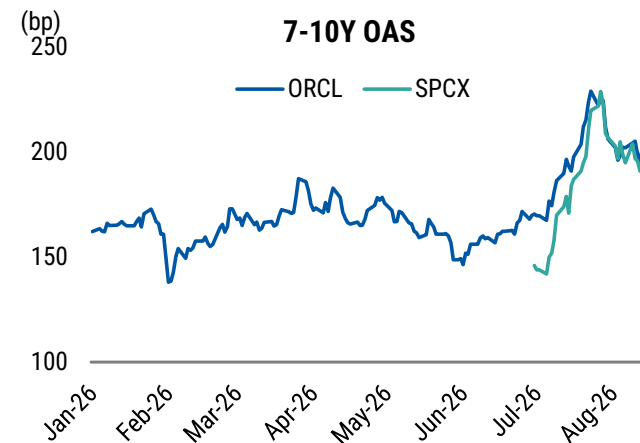
Source: Bloomberg, Morgan Stanley Research

Exhibit 27: NVDA and AVGO 7-10Y bonds trade in line with HQ Hyperscaler bonds...



Source: ICE, Morgan Stanley Research

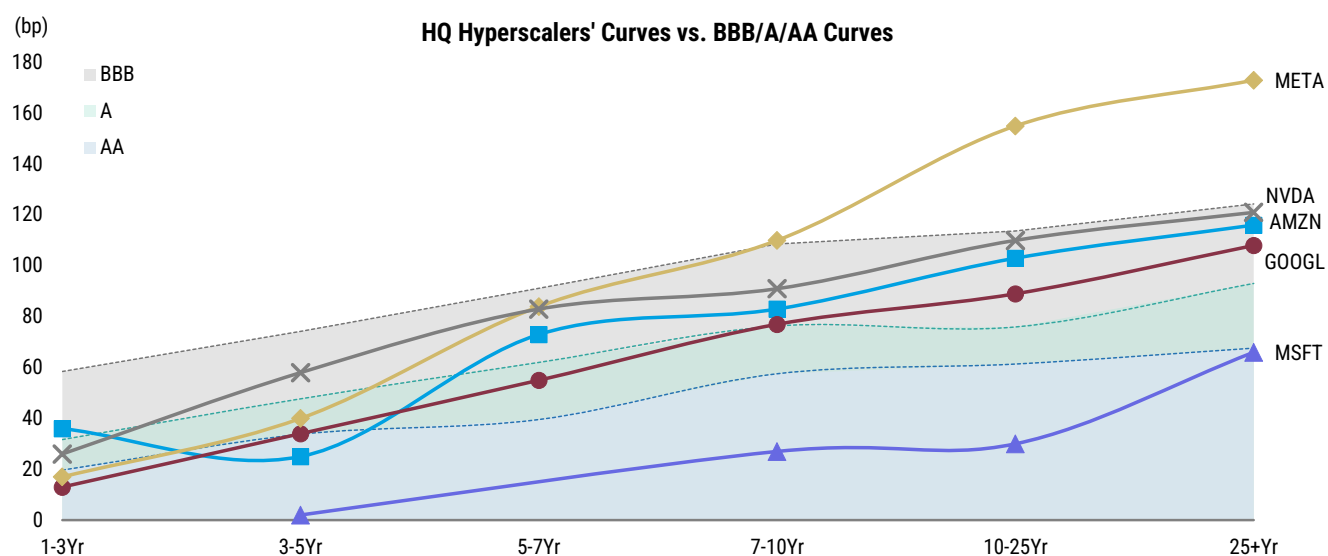
Exhibit 28: ...with BBB AI names tracking similar moves, but trading much wider



Source: ICE, Morgan Stanley Research

Exhibit 29:

META long-end bonds trading closest to par are up to 50bp wider than the BBB curve



Source: ICE, Morgan Stanley Research. Data as of 8/17/2026. Note: The Ratings Curve excludes Financials, Utilities and HQ Hyperscalers (+ORCL SPCX & NVDA). Issuer spreads are derived from bonds closest to par within each tenor bucket.

Exhibit 30: IG AI-Related Tracker - Unsecured Corporate Spreads Tracker

Security Name	Issue Date	Ratings (S/M/F)			Amt Out (\$mn)	Current Level	Spreads as of								Spread Change (bp)			
							7/31/26	6/30/26	5/29/26	4/30/26	3/31/26	2/27/26	1/30/26	1/5/26	MTD	QTD	Since 1Q26	YTD
MSFT 3.45 08/08/36	8/8/2016	AAA	Aaa	WD	1,749	38	45	33	26	37	36	43	28	32	(7)	+5	+2	+6
MSFT 2.921 03/17/52	3/17/2021	AAA	Aaa	WD	6,250	76	79	69	56	59	65	64	53	54	(3)	+8	+11	+23
NVDA 4 1/2 06/15/31	6/18/2026	AA	Aa1	NR	4,000	57	55	36	-	-	-	-	-	-	+2	+21	-	-
NVDA 4.95 06/15/36	6/18/2026	AA	Aa1	NR	4,000	86	86	62	-	-	-	-	-	-	+0	+24	-	-
NVDA 5 5/8 06/15/56	6/18/2026	AA	Aa1	NR	3,500	117	106	74	-	-	-	-	-	-	+11	+42	-	-
GOOGL 4.1 02/15/31	2/13/2026	AA+	Aa2	NR	3,000	46	41	30	26	33	35	41	-	-	+5	+16	+11	-
GOOGL 4.7 11/15/35	11/6/2025	AA+	Aa2	NR	3,500	72	62	51	48	57	57	56	53	54	+10	+21	+15	+18
GOOGL 5.45 11/15/55	11/6/2025	AA+	Aa2	NR	4,000	108	104	79	69	78	77	85	74	74	+4	+29	+31	+34
AMZN 4 1/4 03/13/31	3/13/2026	AA	A1	AA-	5,000	52	54	39	36	42	46	-	-	-	(2)	+13	+6	-
AMZN 4.65 11/20/35	11/20/2025	AA	A1	AA-	3,500	71	71	61	49	63	61	64	54	54	+0	+10	+10	+17
AMZN 5.45 11/20/55	11/20/2025	AA	A1	AA-	3,000	110	112	93	78	86	85	94	70	78	(2)	+17	+25	+33
META 4.55 05/15/31	5/4/2026	AA-	Aa3	NR	3,000	68	73	47	47	53	-	-	-	-	(5)	+21	-	-
META 4 7/8 11/15/35	11/3/2025	AA-	Aa3	NR	6,500	107	100	78	73	82	80	76	75	74	+7	+29	+27	+33
META 5 5/8 11/15/55	11/3/2025	AA-	Aa3	NR	6,500	171	170	141	124	125	117	113	106	109	+1	+30	+54	+63
AVGO 2.45 02/15/31	10/17/2025	A-	A3	A-	2,746	85	74	53	49	50	64	62	54	58	+11	+32	+21	+27
AVGO 4.8 02/15/36	9/29/2025	A-	A3	A-	2,250	122	108	75	70	72	78	82	67	77	+14	+47	+44	+45
AVGO 5.7 01/15/56	1/13/2026	A-	A3	A-	1,250	126	129	83	72	75	82	87	75	-	(3)	+43	+44	-
SPCX 5.35 07/15/31	6/26/2026	BBB	Baa1	BBB+	7,000	141	153	117	-	-	-	-	-	-	(12)	+24	-	-
SPCX 5 7/8 07/15/36	6/26/2026	BBB	Baa1	BBB+	6,000	195	209	160	-	-	-	-	-	-	(14)	+35	-	-
SPCX 6.65 07/15/56	6/26/2026	BBB	Baa1	BBB+	3,500	234	243	201	-	-	-	-	-	-	(9)	+33	-	-
ORCL 4.95 02/04/31	2/4/2026	BBB-	Baa2	BBB	3,500	157	163	123	119	142	149	145	-	-	(6)	+34	+8	-
ORCL 5.2 09/26/35	9/26/2025	BBB-	Baa2	BBB	4,000	197	198	167	141	177	177	173	154	160	(1)	+31	+20	+38
ORCL 5.95 09/26/55	9/26/2025	BBB-	Baa2	BBB	3,500	252	241	229	203	231	237	232	200	201	+11	+23	+15	+52
5s10s Curves						Current	7/31/26	6/30/26	5/29/26	4/30/26	3/31/26	2/27/26	1/30/26	1/5/26	MTD	QTD	Since 1Q26	YTD
NVDA (4.95 '36 - 4 '31)						30	31	26	-	-	-	-	-	-	(1)	+4	-	-
GOOGL (4.7 '35 - 4.1 '31)						26	21	21	22	24	22	15	-	-	+5	+5	+4	-
AMZN (4.65 '35 - 4.25 '31)						19	17	22	13	21	15	-	-	-	+2	(3)	+4	-
META (4.875 '35 - 4.55 '31)						39	27	31	26	29	-	-	-	-	+12	+8	-	-
AVGO (4.8 '36 - 2.45 '31)						37	34	22	21	22	14	20	13	19	+3	+15	+23	+18
SPCX (5.875 '36 - 5.35 '31)						54	56	43	-	-	-	-	-	-	(2)	+11	-	-
ORCL (5.2 '35 - 4.95 '31)						40	35	44	22	35	28	28	-	-	+5	(3)	+12	-
10s30s Curves						Current	7/31/26	6/30/26	5/29/26	4/30/26	3/31/26	2/27/26	1/30/26	1/5/26	MTD	QTD	Since 1Q26	YTD
MSFT (2.921 '52 - 3.45 '36)						38	34	36	30	22	29	21	25	22	+4	+3	+9	+16
NVDA (5.625 '56 - 4.95 '36)						60	51	38	-	-	-	-	-	-	+9	+22	-	-
GOOGL (5.45 '55 - 4.7 '35)						62	63	49	43	45	42	44	-	-	(1)	+13	+20	-
AMZN (5.45 '55 - 4.65 '35)						58	58	54	42	44	39	-	-	-	+0	+4	+19	-
META (5.625 '55 - 4.875 '35)						103	97	94	77	72	-	-	-	-	+6	+9	-	-
AVGO (5.7 '56 - 4.8 '36)						40	55	30	23	25	18	25	21	-	(15)	+10	+22	-
SPCX (6.65 '56 - 5.875 '36)						93	90	84	-	-	-	-	-	-	+3	+9	-	-
ORCL (5.95 '55 - 5.2 '35)						95	78	106	84	89	88	87	-	-	+17	(11)	+7	-

Source: Bloomberg, Morgan Stanley Research. Levels reflect spread to benchmark and as of 8/19/26.

Hyperscaler CDS Performance: We Like Buying CDS Protection

This remains our preferred way of playing the AI story in the credit derivatives market. Liquidity across hyperscaler CDS improved meaningfully since their inclusion in the CDX (March roll), and given the increasing AI-related risk concentration in small set of names, we expect increasing needs for non-economic hedging to manage counterparty exposure (even in a benign scenario in which AI investments are profitable and therefore investments continue). Key risks to this trade: lower issuance vs. estimates, as well as faster-than-expected monetization results in stronger fundamentals, could result in CDS spreads tightening back.

Exhibit 31: AI basket is now trading ~40bp wider to CDX IG

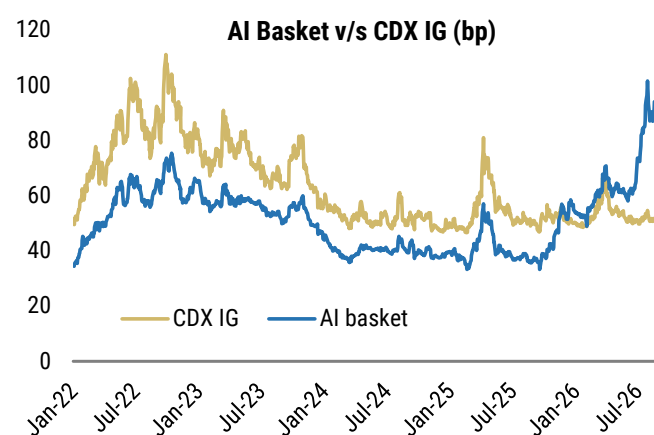


Exhibit 32: Hyperscaler CDS volume went up again with the recent sell-off

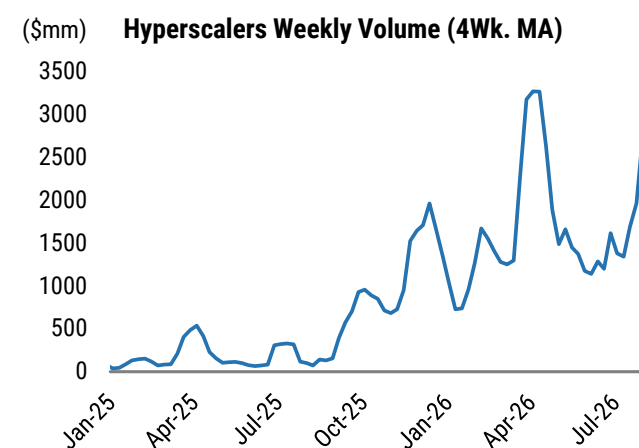


Exhibit 33: MSFT 5Y CDS is the only one that still trades tighter to CDX IG

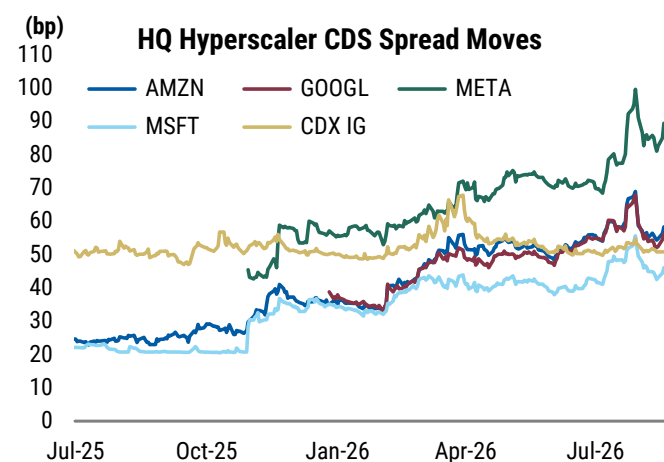
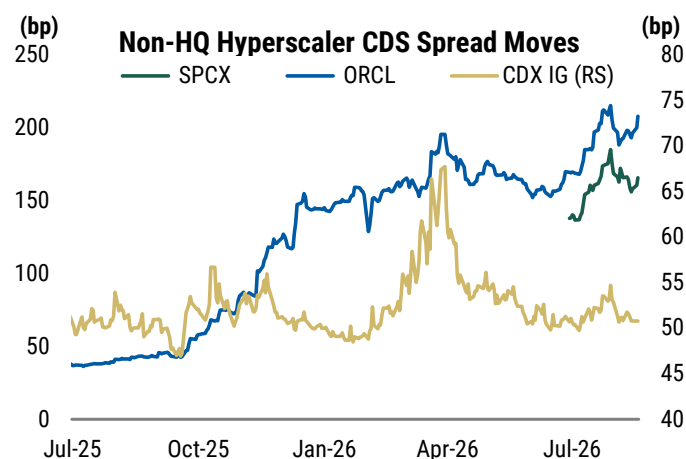


Exhibit 34: We're now tracking ORCL 5Y CDS vs. SPCX 5Y CDS



Collateral-Backed Risk: Data Center and Chip Financing Bonds Trade at a Discount to Unsecured Bonds of Tenants/Guarantors

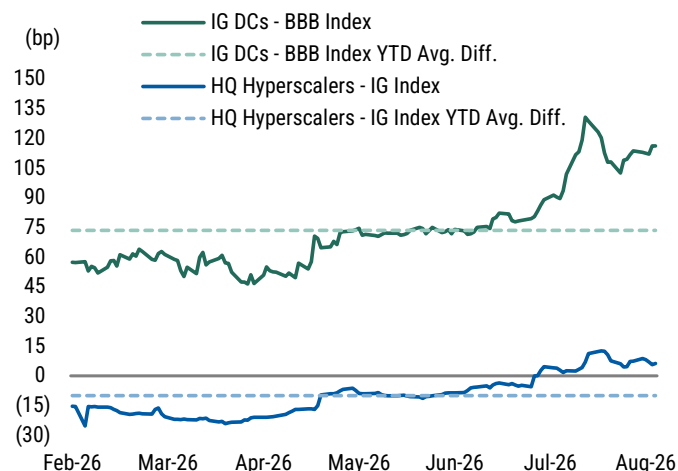
We favor collateral-backed risk ([Exhibit 35](#)) over unsecured corporate risk. For data center bonds, we see prospects of lower LTV once construction is completed and think near-term supply could moderate given increasing political push back and power bottlenecks. We favor deals with high-quality tenants, residual value guarantees, full amortization, and documentation that allows for rent commencement as soon as shells are delivered (less dependent on power availability). As for chip financing, we think the structures offer diversification from long-tenor asset risk with fully amortizing structures and residual value guarantees.

Exhibit 35: Collateral-backed debt is now >\$100bn in IG public/private markets

Issue	Amt. Out. (\$bn)	Market	Ratings (S/M/F)			Sprd to Bench (bps)	
						Issue	Current
Data Center Bonds							
RPLDCI 6.581 05/30/49	27	144A Priv. Plac.*	A+	NR	NR	254	234
QTSQST 5.7 04/15/36	5	144A Priv. Plac.	NR	Baa2	NR	138	239
RDMICH 7 1/2 03/30/45	14	144A Priv. Plac.	NR	Baa3	BBB	329	346
HUTRBA 6.192 11/15/42	3	144A Priv. Plac.	(P)BBB-	NR	BBB-	185	183
HUTBPA 6.129 11/30/42	4	144A Priv. Plac.	NR	Baa2	NR	165	195
TXEDCI 7.534 11/30/48	13	144A Priv. Plac.*	A+	NR	AA-	288	256
QTSODY 6 5/8 09/01/31	4	144A Priv. Plac.	NR	NR	BBB-(EXP)	286	259
Total DC Financing	70						

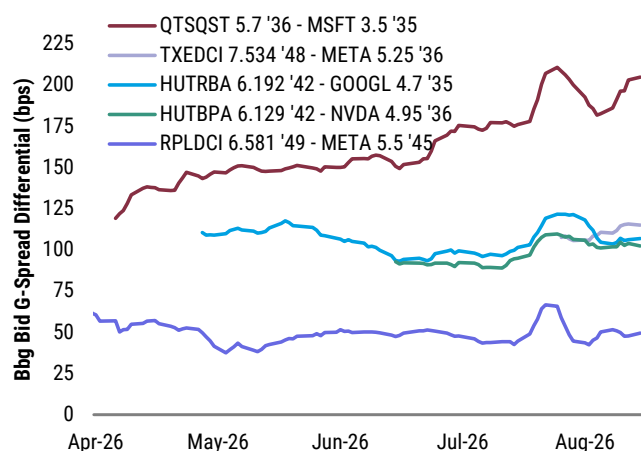
Source: Bloomberg, Morgan Stanley Corporate Trading Desk, Morgan Stanley Research. Note: * denotes the Beignet and Sopapilla transactions. The total of >\$100bn includes \$35bn of chip financing associated to AVGO.

Exhibit 36: IG DC bonds trade ~100bp wide to the IG index



Source: Bloomberg, Morgan Stanley Research

Exhibit 37: IG DC bonds trade 50-200bp back of their tenants' unsec'd bonds; deal structure drives dispersion



Source: Bloomberg, Morgan Stanley Research

Investment Grade Reverse Yankee: Hyperscalers Have Now Tapped 6 Non-USD Currencies in 2026

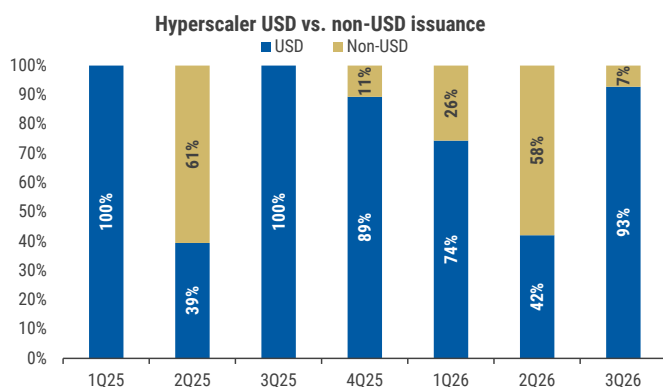
Hyperscalers Diversified Their Funding Base And Tapped Numerous Currencies This Year

Of the \$200bn+ of bonds issued by hyperscalers this year, more than a quarter has been financed in non-USD markets. Indeed, the majority of issuance was in non-domestic currency in 2Q26. In contrast to 2025, this year saw a significant broadening out of currencies, with only 40% of non-USD supply coming in EUR this year. Issuers further diversified their funding towards five other currencies, including CAD, GBP, CHF, AUD and JPY (Exhibit 38).

And while the ability of markets to absorb longer dated bonds varies across currencies, the hyperscaler deals were more skewed towards the long-end than the respective currency bond markets in nearly all markets except the recent AUD deal (Exhibit 40).

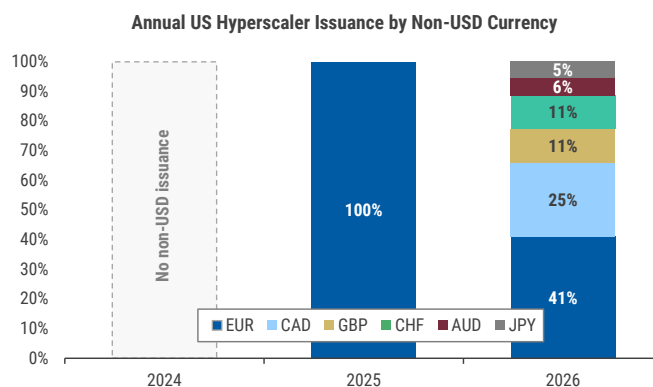
Given their relatively large deal sizes (Exhibit 42 and Exhibit 43), these issuers have already become some of the largest issuers in many of the local IG benchmarks (Exhibit 40). However, since only two issuers (Alphabet and Amazon) have utilised the non-dollar markets so far this year, sector concentration remains less pronounced outside the USD market for now.

Exhibit 38: 2Q26 saw a majority of hyperscaler supply coming in non-USD currencies



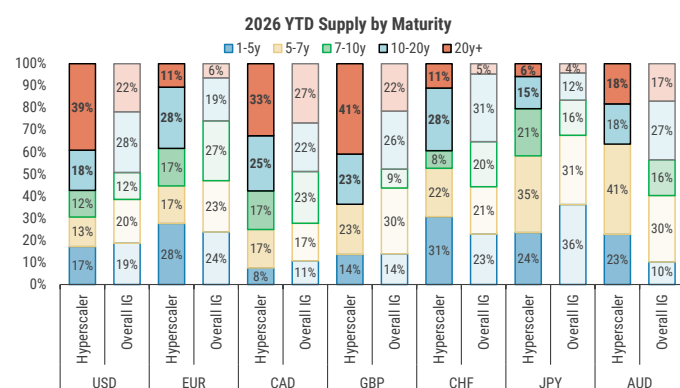
Source: Bloomberg, Bond Radar, Morgan Stanley Research

Exhibit 39: The pace of non-USD issuance from hyperscalers has proven resilient



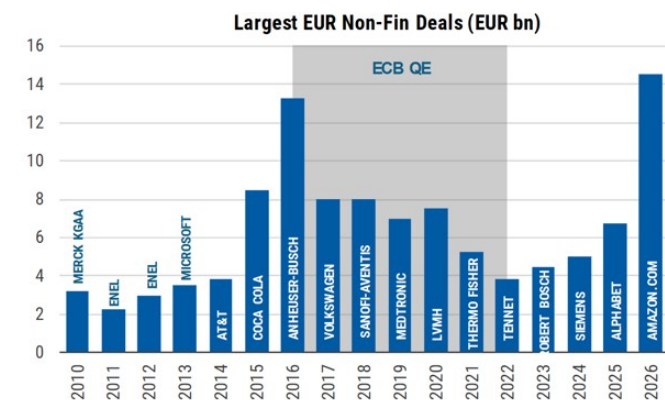
Source: Bloomberg, Bond Radar, Morgan Stanley Research

Exhibit 40: Hyperscaler supply is skewed towards longer maturities, both outright and relative to the overall IG benchmark



Source: Bloomberg, Bond Radar, Morgan Stanley Research

Exhibit 42: Hyperscalers have shown EUR credit's ability to absorb QE-era-sized jumbo deals



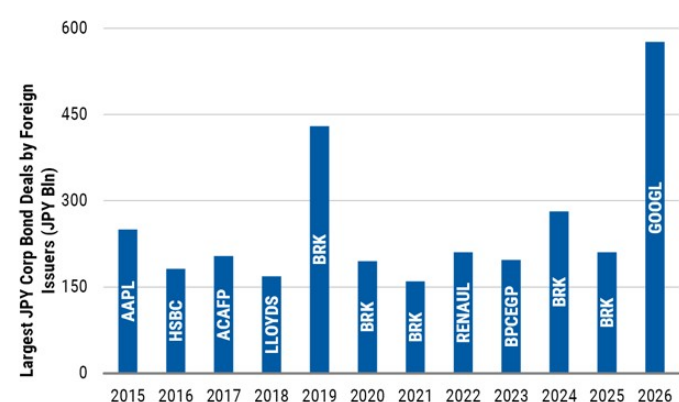
Source: ICE BofA, Morgan Stanley Research

Exhibit 41: Hyperscalers are already heavily represented in non-USD IG credit indices

				Weight in IG Non-Fin Index							Rank in IG Non-Fin Index						
				USD	EUR	GBP	CHF	CAD	JPY	AUD	USD	EUR	GBP	CHF	CAD	JPY	AUD
Hyperscalers	ORCL	BBB2	Technology & Electronics	1.5%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2	-	-	-	-	-	-
	AMZN	AA3	Retail	1.7%	0.6%	0.0%	4.4%	3.4%	0.0%	0.0%	1	29	-	5	5	-	-
	META	AA3	Media	1.2%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	4	-	-	-	-	-	-
	MSFT	AAA	Technology & Electronics	0.4%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	67	249	-	-	-	-	-
	GOOGL	AA2	Media	1.1%	1.1%	2.5%	4.7%	2.1%	8.1%	4.8%	6	9	2	4	13	5	1
	Total			5.8%	1.8%	2.5%	9.0%	5.5%	8.1%	4.8%							
AI Adj.	AVGO	A3	Technology & Electronics	0.8%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	14	-	-	-	-	-	-
	NVDA	A3	Technolov & Electronics	0.5%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	42	-	-	-	-	-	-

Source: Bloomberg, Bond Radar, Morgan Stanley Research; Note: Includes hyperscaler deals from August that are index-eligible but have not yet entered the latest index.

Exhibit 43: Alphabet's JPY deal was the biggest yen bond on record sold by a foreign issuer



Source: ICE BofA, Morgan Stanley Research

Reverse Yankee Spread Performance

The performance of hyperscaler new issues this year has been mixed, with many deals seeing spread widening in the first 20 days of trading ([Exhibit 44](#)).

Indeed, hyperscalers' EUR bonds have underperformed the broader market particularly since early June, with the belly of the curve now trading closer to the A rated index than AA ([Exhibit 45](#)). The underperformance was particularly pronounced in the long-end, with these issuers' curves screening particularly steep compared to the overall market ([Exhibit 46](#)).

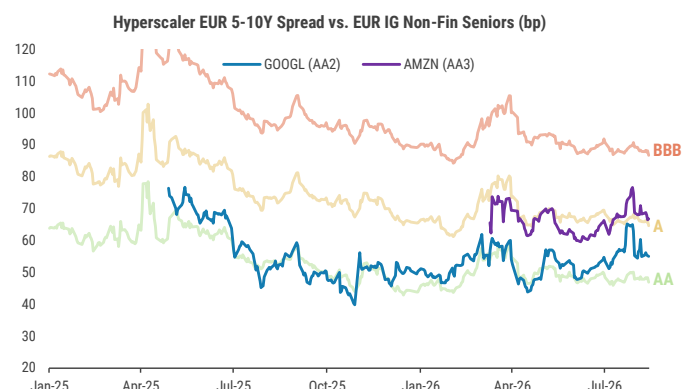
However, while hyperscalers in EUR have underperformed within the EUR benchmark, they have outperformed their USD counterparts. On a cross-currency adjusted spread basis, EUR hyperscaler bonds currently trade tight to their USD bonds ([Exhibit 47](#)).

Exhibit 44: Hyperscaler new issues' performance has been mixed the month following pricing...

Date	Ccy	Ticker	Avg. New Issue Spread (bp)	Spread	NIC	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Feb-26	USD	ORCL	147	-21		-4	-1	+7	+11	+8	+12	+12	+13	+15		+18	+15	+13	+14	+23	+25	+20	+25	+37	+36
Feb-26	USD	GOOGL	63	9		-1	-3	-3	-2		-2	-3	-3	-4		-2	-3	-2	+1	-1	-2	-4	-2	-1	-1
Feb-26	CHF	GOOGL	70			+1	+2	+2	+2	+2	+0	-0	+1	+0	+1	+2	+1	+2	+2	+2	+1	+4	-0	+10	+1
Feb-26	GBP	GOOGL	67			-1	-1	+1	+2	+4	+1	-0	-2	-1	-1	+1	+3	+6	+8	+1	+1	+6	+7	+4	
Feb-26	EUR	NTT	93	0		+5	+9	+2	+11	+10	+2	+6	+12	+3	+2	+4	+4	+11	+9	+1	+6	-5	+16	+8	+6
Feb-26	GBP	NTT	85			+1	+4	+5	+8	+3	+1	+4	+7	+3	+4	+6	+7	+8	+9	+8	+12	+13	+10	+10	+10
Mar-26	USD	AMZN	79	15		-2	-3	-2	-3	-5	-5	-6	-7	-8	-10	-11	-10	-7	-7	-10	-14	-14	-16	-13	
Mar-26	EUR	AMZN	93			+2	+4	+1	+0	+3	+3	+8	+5	+0	-0	+0	+2	+1	-2	-2			-1	-5	
May-26	EUR	GOOGL	91	8		+0	+1	+4	+3	+2	-0	+0	-2	-1	-10	-1	-4	-2	-1	-1	+0	-1	-2	-7	
May-26	CAD	GOOGL	74			-5	-4	-3	-3	-4	-3	-4	-4		-4	-6	-5	-6	-6	-6	-7	-6	-7	-8	
May-26	CHF	AMZN	89			+1	-0	+3	+1	+1	+2	+2		+2	+2	+2	+2	+2	+3	+3	+2	+3	+2		
May-26	JPY	GOOGL	79			-0	-2	+2	-0	+1	+2	+2	+2	+4	+4	+1	+5	+3	+4	+0	+1	+4	+3	+2	
Jun-26	CAD	AMZN	70			-0	-0	-1	-1	-2	-1	-1	+1	+2	+3	+4	+4	+4	+5	+4	+4		+3	+3	+3
Jun-26	USD	DELL	76	-7		-1	-1	+1	+2	+3		+3	+5	+7	+8	+10	+8	+9	+8	+6		+6	+10	+8	+8
Jun-26	USD	NVDA	43	9		+2	+3	+4		+4	+6	+7	+7	+6	+6	+6	+5		+5	+11	+10	+12	+15	+14	
Jun-26	USD	SPCX	144			+9	+14	+20	+17	+16	+18	+17		+15	+24	+24	+30	+42	+45	+52	+46	+44	+46	+49	+54
Jun-26	USD	NTT	83	1		-1	-1	-0	+0	-1	-2		-2	-3	-3	-4	-4	-5	-5	-4	-4	-4	-4	-4	-4
Jun-26	EUR	NTT	91	13		-5	-3	-4	-1	-2	-5	-7	-8	-10	-7	-7	-10	-4	-7	-4	-8	-6	-8	-5	-5
Jun-26	GBP	NTT	80			-3	-3	-3	-3	-4	-4	-3	-4	-3	-2	-3	-3	-3	-2	-1	-2	-2	-2	-2	-2
Jul-26	USD	AMZN	83	21		-5	-2	-1	-0	+2	-2	-4	-2	-3	-1	-1	+3	+7	+7	+11	+10	+5	+1	-1	-4
Jul-26	EUR	T	121	14		+0	+1	-3	-4	-5	-2	-5	-4	-5	-5	-6	-7	-6							
Aug-26	USD	GOOGL	81	21		-5	-5	-2	-6	-7	-5	-5													

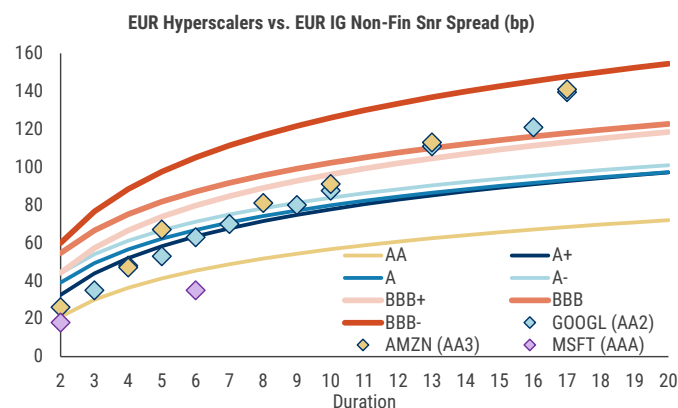
Source: Bloomberg, Bond Radar, Morgan Stanley Research

Exhibit 45: ...and in secondary, hyperscalers have also underperformed broader EUR credit



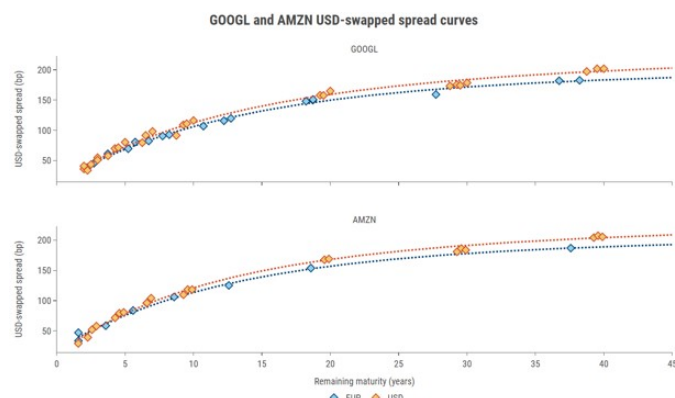
Source: Bloomberg, IHS Markit, Morgan Stanley Research; Note: Hyperscaler spreads reflect the simple average of mid Z-spread among bonds with maturities from 5-10 years.

Exhibit 46: Hyperscaler credit curves are steep relative to their rating cohort



Source: ICE BofA, Morgan Stanley Research

Exhibit 47: On a cross-currency adjusted spread basis, hyperscaler EUR bonds trade within the USD curve



Source: Bloomberg, Morgan Stanley Research

We summarize relevant book stats and price/spread performance for hyperscaler issuance across European currencies below.

Exhibit 48: Reverse yankees – performance at a glance

Pricing Date	Ccy of Issue	Security	Amount Issued (local ccy mns)	Book Stats			Price				Spreads As Of (bp)			
				New Issue Premium (bp)	Bid-Cover (x)	IPT-Launch Spread Compression (bp)	19-Aug-26	Δ Since Issue	Δ YTD	Δ MTD	19-Aug-26	Δ Since Issue	Δ YTD	Δ MTD
29-Apr-25	EUR	GOOGL 2 1/2 05/06/29	1,500	-	4.2x	-33	98	-1.5	-1.4	-0.8	22	-30	-4	-3
		GOOGL 3 05/06/33	1,500	-	4.3x	-33	96	-3.9	-2.9	-2.6	63	-9	+11	+7
		GOOGL 3 3/8 05/06/37	1,250	-	4.2x	-30	93	-5.6	-3.7	-3.6	89	-6	+17	+8
		GOOGL 3 7/8 05/06/45	1,250	-	4.2x	-30	89	-11.1	-7.0	-5.8	140	+10	+38	+17
		GOOGL 4 05/06/54	1,250	-	4.6x	-30	86	-13.1	-7.2	-7.0	160	-0	+34	+18
03-Nov-25	EUR	GOOGL 2 3/8 11/06/28	1,000	-	5.7x	-35	98	-1.4	-1.2	-0.6	12	-13	-8	-5
		GOOGL 2 7/8 11/06/31	1,000	18	-	-33	97	-3.0	-2.2	-1.8	46	-1	+2	+1
		GOOGL 3 1/8 11/06/34	1,000	8	-	-33	95	-4.6	-3.0	-2.9	71	+9	+10	+6
		GOOGL 3 1/2 11/06/38	1,000	6	-	-33	92	-6.3	-4.2	-4.1	99	+17	+19	+10
		GOOGL 4 11/06/44	1,250	18	-	-30	91	-8.2	-6.5	-6.4	136	+21	+32	+21
10-Feb-26	GBP	GOOGL 4 3/8 11/06/64	1,250	-	-	-32	87	-11.5	-8.8	-7.4	192	+34	+45	+17
		GOOGL 4 1/8 02/13/29	750	-	4.9x	-	98	-1.5	-	-0.7	31	-14	-	+8
		GOOGL 4 5/8 11/13/32	1,250	-	3.8x	-	97	-3.1	-	-1.9	49	-6	-	+7
		GOOGL 5 1/2 11/13/41	1,250	-	3.5x	-	92	-6.8	-	-4.6	75	+20	-	+17
		GOOGL 5 7/8 02/13/58	1,250	-	5.7x	-	89	-9.9	-	-6.5	86	+26	-	+19
10-Feb-26	CHF	GOOGL 6 1/8 02/13/2126	1,000	-	9.8x	-	89	-10.4	-	-5.6	140	+20	-	+13
		GOOGL 0.427 03/01/29	905	-	-	-10	99	-0.5	-	-0.5	45	+5	-	+14
		GOOGL 0.89 03/03/32	700	-	-	-5	99	-1.0	-	-1.2	66	+6	-	+6
		GOOGL 1.2525 03/03/36	575	-	-	-5	99	-1.5	-	-2.1	80	+10	-	+9
		GOOGL 1.5823 03/01/41	455	-	-	-3	97	-3.2	-	-3.7	101	+19	-	+15
11-Mar-26	EUR	GOOGL 1.8675 03/03/51	420	-	-	-2	97	-3.0	-	-4.2	111	+13	-	+11
		AMZN Float 03/16/28	1,750	-	-	-	100	+0.1	-	+0.0	-	-	-	-
		AMZN 2.8 03/16/28	1,250	-	1.8x	-25	100	-0.4	-	-0.4	9	-21	-	-6
		AMZN 3.1 03/16/30	2,000	-	2.1x	-27	99	-1.0	-	-1.2	37	-11	-	-1
		AMZN 3.35 03/16/32	2,250	-	2.1x	-27	98	-1.7	-	-1.9	61	-2	-	+0
05-May-26	EUR	AMZN 3.7 03/16/35	2,500	-	2.3x	-27	97	-2.4	-	-3.3	86	+3	-	+8
		AMZN 4.05 03/16/39	2,250	-	1.6x	-27	97	-2.8	-	-4.0	106	+3	-	+5
		AMZN 4.45 03/16/45	1,250	-	2.0x	-28	96	-3.8	-	-5.6	141	+6	-	+13
		AMZN 4.85 03/16/64	1,250	-	2.1x	-25	94	-5.7	-	-8.0	193	+5	-	+17
		GOOGL 3.2 05/11/30	1,500	7	1.8x	-33	99	-0.8	-	-1.4	38	+3	-	+3
12-May-26	CHF	GOOGL 3.45 05/11/32	1,750	11	1.9x	-28	99	-1.1	-	-2.1	57	+2	-	+2
		GOOGL 3 5/8 05/11/34	1,500	4	2.0x	-32	98	-1.6	-	-2.9	70	+7	-	+6
		GOOGL 4.1 05/11/39	1,750	7	1.9x	-32	98	-2.3	-	-4.3	103	+10	-	+9
		GOOGL 4 1/2 05/11/45	1,250	16	2.7x	-30	96	-3.5	-	-5.9	140	+15	-	+15
		GOOGL 4.8 05/11/63	1,250	11	2.1x	-30	94	-5.4	-	-8.0	189	+14	-	+20
		AMZN 0.835 05/28/29	900	-	-	+3	100	+0.4	-	-0.2	49	-6	-	-5
		AMZN 1.2 05/28/32	585	-	-	+3	100	-0.1	-	-1.1	80	+5	-	+5
		AMZN 1.4475 05/28/35	475	-	-	+3	100	-0.3	-	-1.6	90	+5	-	+4
		AMZN 1.675 05/28/38	390	-	-	+3	99	-1.0	-	-2.6	105	+10	-	+9
		AMZN 1.9389 05/25/44	245	-	-	+1	99	-1.2	-	-2.8	117	+9	-	+7
		AMZN 2.0825 05/25/51	225	-	-	+1	98	-1.7	-	-3.7	127	+9	-	+7

Source: Bloomberg, Bond Radar, Morgan Stanley Research estimates; Note: Bid Z-spread is shown for EUR and CHF bonds, bid G-spread for GBP bonds.

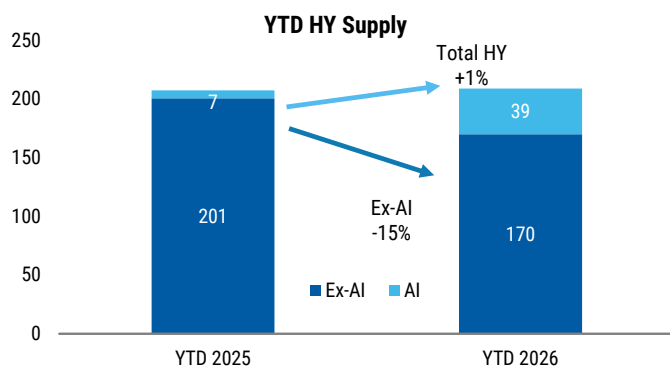
Leveraged Finance: Loan Supply Accelerates; More Sensitivity to Higher Funding Costs

AI-related leveraged finance issuance has accelerated sharply in 2026, with HY issuance reaching ~\$40bn YTD and leveraged loans reaching ~\$10bn, as data center and broader AI financing needs continue to grow. Loan issuance picked up during July and August, with ~\$7bn out of the total ~\$10bn of loan supply. Within HY, data center debt is predominantly secured/project-finance-style (~82%), and AI exposure still represents only ~3% of the overall HY market, suggesting meaningful capacity remains for additional issuance vs. a hypothetical ~15% concentration limit (*lower than what we've seen for other sectors in HY historically*).

However, higher funding costs are becoming an increasingly important constraint for lower-rated borrowers, and could begin to delay or pull new transactions (see our note [A Break from the Heat, For Now](#) for more). At the same time, stabilization provides a natural refinancing path for HY construction debt into ABS/CMBS or potentially IG, while chip financing and other asset-level structures should create additional opportunities for the loan and private credit markets.

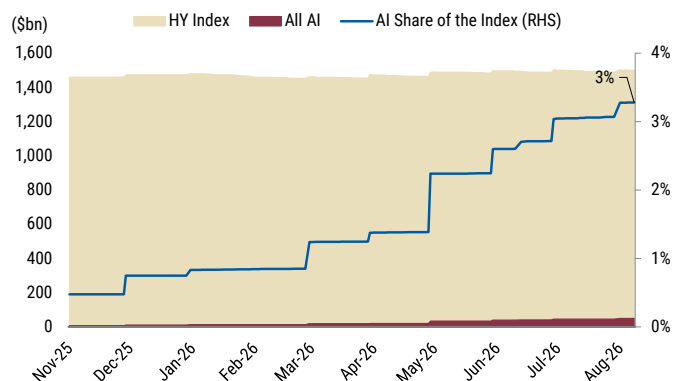
High Yield: Data Center Deals Continue to Drive HY YTD Supply

Exhibit 49: Ex-AI, HY YTD supply is -15% y/y



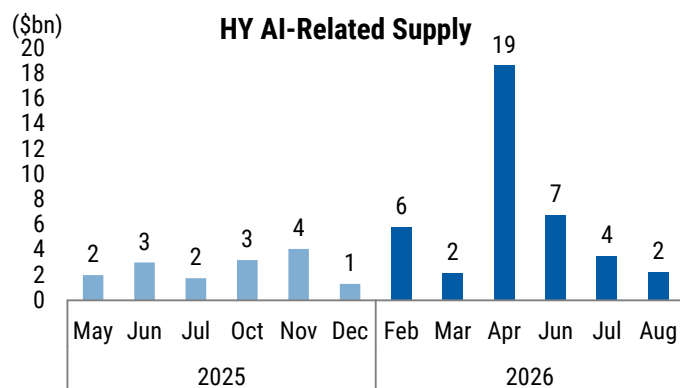
Source: Pitchbook LCD, Morgan Stanley Research; As of 8/13/2026

Exhibit 50: AI related debt accounts for 3% of the HY index, up from <1% last year



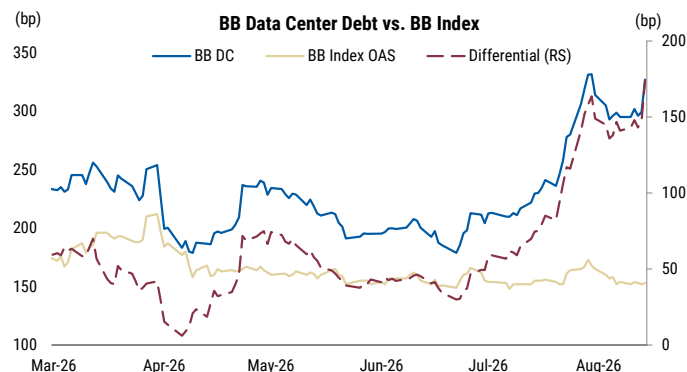
Source: ICE BAML, Morgan Stanley Research

Exhibit 51: So far this August we have seen only one HY AI-related deal, down from the rush in April and June



Source: Pitchbook, Morgan Stanley Research

Exhibit 52: BB-rated HY data center backed deals are now trading 180bp back from the BB index, vs. 80bp on YTD average



Source: Bloomberg, Morgan Stanley Research

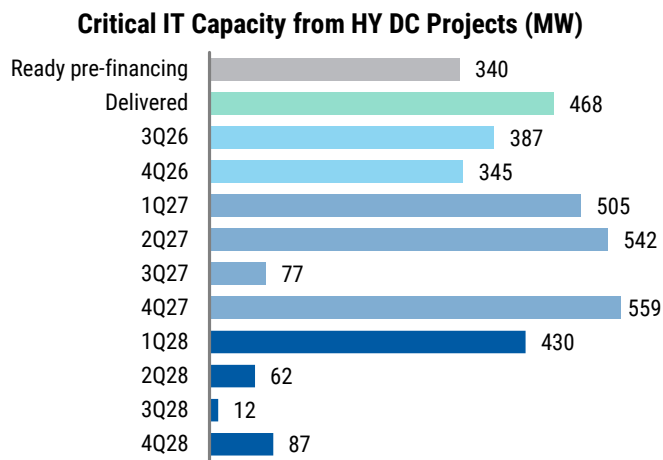
HY Data Center Bonds: We're Seeing More Dispersion After The Sell-Off

Exhibit 53: There are now 18 project finance style bonds in the HY market spanning 7 tenants.

Issue	Amt Out (\$bn)	Pricing Date	YTC (%)	S&P	Moody's	Fitch	Location	Critical IT MW	Tenant	Add'l Credit Support	First Call Date	Final Delivery Date
BLKPRL 6.125 '31	2.00	2/4/2026	7.9%	NR	Ba2	BB-	Wink, TX	216	Amazon	-	2/15/2028	2/1/2027
STNGRY 6.000 '31	0.81	6/8/2026	8.8%	NR	Ba2	BB-	Andrews, TX	70	Amazon	-	6/15/2028	4/1/2027
APLD 9.250 '30	2.35	11/13/2025	8.2%	BB-	NR	BB-	Ellendale, ND	250	CoreWeave	-	12/15/2027	10/31/2026
CORZ 7.750 '31	3.30	4/22/2026	11.5%	BB	NR	BB-	TX,GA,OK, NC	589	CoreWeave	-	5/15/2028	6/30/2027
EDGCOM 7.500 '31	1.30	4/21/2026	14.2%	BB+	NR	BB-	Chicago/Atlanta	114	CoreWeave/Alibaba	-	4/30/2028	12/1/2027
ELKGVP 7.500 '31	0.90	6/2/2026	12.0%	BB-	Ba3	BB-	Chicago, IL	72	CoreWeave	-	6/15/2028	1/1/2027
ELNFOR 7.000 '31	1.59	6/9/2026	11.8%	BB-	NR	BB-	Ellendale, ND	150	CoreWeave	-	6/15/2028	6/1/2027
GALAXY 9.875 '31	3.51	7/28/2026	12.8%	BB-	NR	BB-	Lubbock, TX	260	CoreWeave	-	8/1/2028	3/31/2028
CIFR 7.125 '30	1.73	11/5/2025	8.4%	NR	Ba3	BB-	Colorado City, TX	207	Fluidstack	\$1.733bn Google backstop	11/15/2027	9/30/2026
FLASHC 7.250 '30	1.30	12/18/2025	9.3%	NR	Ba3	NR	Abernathy, TX	168	Fluidstack	\$1.3bn Google backstop	12/31/2027	9/30/2026
MERIDI 6.250 '31	5.70	4/16/2026	11.1%	(P)BB-	Ba2	BB	Sullivan County, IN	430	Fluidstack	Google lease guarantee	4/30/2028	2/1/2028
WULF 7.750 '30	3.20	10/16/2025	7.4%	BB	Ba2	BB	Lake Mariner, NY	438	Fluidstack	\$3.2bn Google backstop	10/15/2027	12/31/2026
ZENARC 8.875 '31	2.25	8/13/2026	13.0%	(P)BB	Ba2	NR	Oklahoma	149	Jane Street	-	8/31/2028	10/1/2028
TRACTC 5.875 '31	3.80	2/13/2026	13.7%	(P)BB+	Ba1	BB	Storey County, NV	200	NVIDIA	-	3/1/2028	6/1/2028
TRACTD 6.500 '31	4.59	4/28/2026	12.9%	(P)BB	Ba1	BB	Storey County, NV	200	NVIDIA	-	5/1/2028	4/4/2028
PFORGE 6.750 '31	2.15	3/3/2026	10.6%	NR	Ba3	BB-	Harwood, ND	200	Oracle	-	3/15/2028	6/1/2027
YNDRDC 6.875 '31	0.72	6/23/2026	12.6%	NR	Ba3	BB	Loudoun County, VA	48	Oracle	-	6/30/2028	10/1/2028
SECMOS 8.875 '31	1.00	4/30/2026	12.0%	(P)BB+	NR	BB-	Austin, TX	50	SoftBank	Softbank	5/1/2028	6/1/2027
Total:	42.19							3,811				

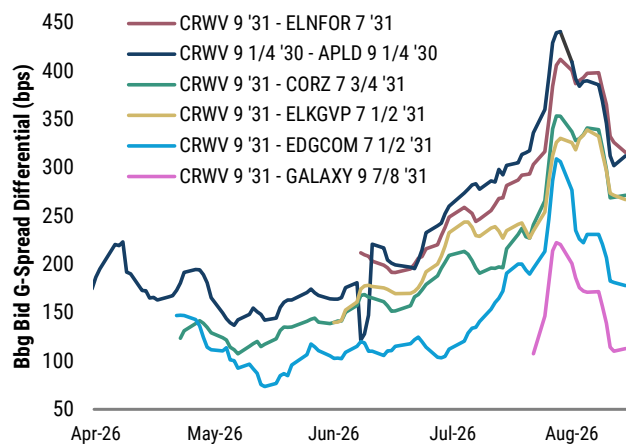
Source: Bloomberg, Morgan Stanley Research; As of 8/20/2026

Exhibit 54: Capacity delivery tracker based on disclosed schedules and public announcements from issuers



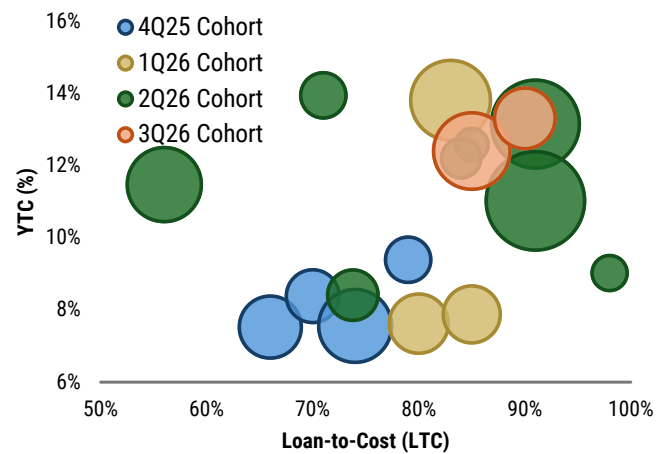
Source: Bloomberg, Moody's, Fitch, S&P, Morgan Stanley Research

Exhibit 56: CRWV unsecured bonds underperformed vs. the HY DC bonds that have CRWV as tenant



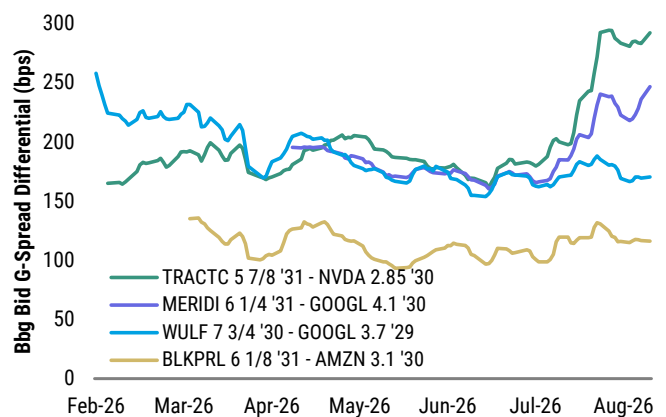
Source: Bloomberg, Morgan Stanley Research

Exhibit 55: LTC went up since Fall 2025; 2Q26 cohort has more dispersion than other cohorts

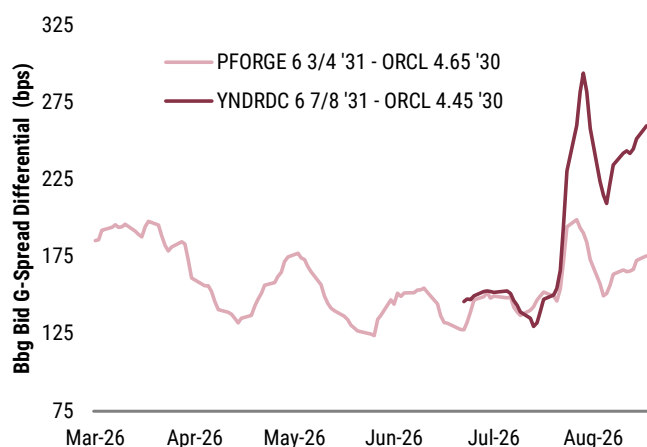


Source: Bloomberg, Moody's, Fitch, S&P, Morgan Stanley Research. Note: bubble size represents amount outstanding for each bond.

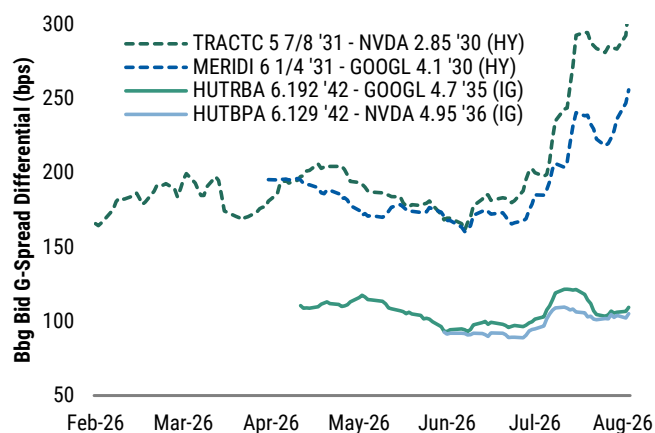
Exhibit 57: HY DC bonds with IG tenants or IG credit support trade 100-250bp back of the IG tenant's unsecd bonds



Source: Bloomberg, Morgan Stanley Research

Exhibit 58: HY DC bonds with ORCL as tenant trade ~175-260bp wide of ORCL unsec'd bonds of similar duration

Source: Bloomberg, Morgan Stanley Research

Exhibit 59: IG vs. HY DC bonds: Wider differential for HY bonds driven by higher construction and refinancing risks

Source: Bloomberg, Morgan Stanley Research

Exhibit 60:

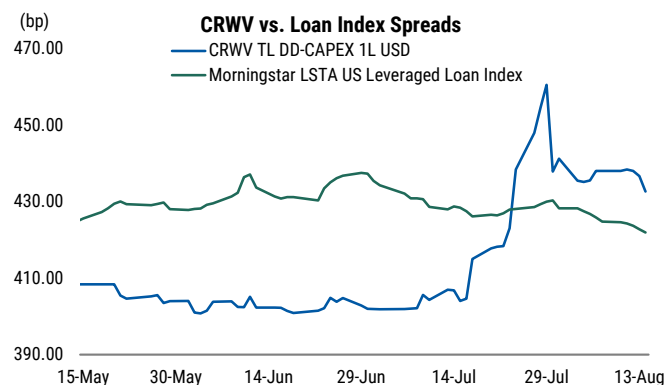
HY Issuer Performance – Debt & Equity

							Spread Change (Equity % Change)								
Type	Entity	Description	Seniority	Price	Yield	Spread	1d	1wk	2wk	1mth	Year to Date (since issuance)	S&P	Moody's	Fitch	Out (bm)
HY	BLACK PEARL COMPUTE LLC	BLKPRL 6 1/8 02/15/31	Secured	100.54	5.95	142	-5	+5	+4	+10	-107	-	Ba2	BB-	2.00
HY	STINGRAY COMPUTE	STNGRY 6 06/15/31	Secured	98.08	6.47	196	-10	+11	+13	+30	+19	-	Ba2	BB-	0.81
HY	APLD COMPUTECO LLC	APLD 9 1/4 12/15/30	Secured	105.45	7.60	246	-12	+14	+28	+45	-346	BB-	-	BB-	2.35
HY	CORE SCIENTIFIC FINANCE	CORZ 7 3/4 05/15/31	Secured	97.77	8.36	387	-11	+27	+19	+17	-7	BB	-	BB-	3.30
HY	EDGED COMPUTE LLC	EDGCOM 7 1/2 04/30/31	Secured	94.28	9.22	478	-20	+30	+3	+77	+140	BB+	-	BB-	1.30
HY	ELK GROVE VILLAGE PROPER	ELKGVP 7 1/2 06/15/31	Secured	96.32	8.55	412	-22	+57	+67	+57	+93	BB-	Ba3	BB-	0.90
HY	APLD COMPUTECO 3 LLC	ELNFOR 7 06/15/31	1st lien	95.68	8.17	379	-2	+81	+87	+57	+111	BB-	-	BB-	1.59
HY	GALAXY HELIOS DATA CNTR	GALAXY 9 7/8 08/01/31	Secured	99.29	10.06	545	-3	+52	+45	-	-9	BB-	-	BB-	3.51
HY	CIPHER COMPUTE LLC	CIFR 7 1/8 11/15/30	1st lien	101.96	6.58	185	-9	+16	+15	+36	-80	-	Ba3	BB-	1.73
HY	FLASH COMPUTE LLC	FLASHC 7 1/4 12/31/30	Secured	101.09	6.87	238	-17	+19	+25	+46	-117	-	Ba3	-	1.30
HY	MERIDIAN ARC HOLDCO LLC	MERIDI 6 1/4 04/30/31	Secured	95.70	7.35	294	-4	+38	+35	+89	+98	BB-	Ba2	BB	5.70
HY	WULF COMPUTE LLC	WULF 7 3/4 10/15/30	Secured	104.09	6.30	168	-5	+12	+13	+17	-125	BB	Ba2	BB	3.20
HY	ZENITH ARC LLC	ZENARC 8 7/8 08/31/31	Secured	96.21	9.85	529	-15	-	-	-	+27	(P)BB	Ba2	-	2.25
HY	SV RNO PROPERTY OWNER 1	TRACTC 5 7/8 03/01/31	Secured	92.19	8.02	358	-12	+45	+46	+142	+166	(P)BB+	Ba1	BB	3.80
HY	PR RNO PROPERTY OWNER 1	TRACTD 6 1/2 05/01/31	Secured	93.40	8.25	386	-5	+45	+47	+155	+143	(P)BB	Ba1	BB	4.59
HY	APLD COMPUTECO 2 LLC	PFORGE 6 3/4 03/15/31	1st lien	97.62	7.44	296	-10	+13	+29	+47	-16	-	Ba3	BB-	2.15
HY	YONDR JK 1 LLC	YNDRDC 6 7/8 06/30/31	1st lien	93.56	8.65	421	-1	+53	+71	+142	+190	-	Ba3	BB	0.72
HY	SE COSMOS LLC	SECMOS 8 7/8 05/01/31	1st lien	99.24	9.07	419	-38	+7	-8	+30	-40	(P)BB+	-	BB-	1.00
HY	COREWEAVE INC	CRWV 9 02/01/31	Sr Unsecured	91.62	11.45	705	+10	+71	-3	+103	-44	B	B1	BB-	1.75
HY	COREWEAVE INC	CRWV 9 1/4 06/01/30	Sr Unsecured	94.61	11.02	663	+27	+59	-17	+83	-95	B	B1	BB-	2.00
HY	COREWEAVE INC	CRWV 9 3/4 10/01/31	Sr Unsecured	91.71	11.96	758	+15	+63	-10	+99	+244	B	B1	BB-	2.75
HY	COREWEAVE INC	CRWV 9 5/8 07/15/32	Sr Unsecured	90.22	11.97	747	+16	+54	+28	+83	+244	B	B1	BB-	1.25
HY	IRON MOUNTAIN INC	IRM 6 1/4 01/15/35	Sr Unsecured	99.17	6.38	143	+2	+6	+10	+12	-1	BB-	Ba3	-	1.50
Equity	APPLIED DIGITAL CORP	APLD Equity	-	28.51	-	-	1.0%	-6.9%	-1.2%	-5.1%	16.3%	-	-	-	-
Equity	CIPHER DIGITAL INC	CIFR Equity	-	15.84	-	-	-0.3%	-4.7%	-13.0%	-30.8%	7.3%	-	-	-	-
Equity	COREWEAVE INC	CRWV Equity	-	91.22	-	-	0.4%	-14.2%	6.9%	14.6%	27.4%	-	-	-	-
Equity	GALAXY DIGITAL INC	GLXY Equity	-	23.04	-	-	5.2%	10.0%	17.9%	-9.3%	3.0%	-	-	-	-
Equity	TERAWULF INC	WULF Equity	-	15.84	-	-	2.2%	-2.9%	-10.1%	-20.3%	37.9%	-	-	-	-
Wavg BB OAS 333															
Total Outstanding														51.44	

Source: Bloomberg, Morgan Stanley Research; As of 8/20/2026. Year to Date refers to year since issuance for bonds issued in 2026.

Lev Loans: Loans Emerge as an Avenue for Chip Financing

Exhibit 61: CRWV's May GPU financing loan now trades ~10bp back from the broader Loans index



Source: Bloomberg, Pitchbook LCD, Morgan Stanley Research

Exhibit 62: We've seen 4 public loan transactions for AI-related assets; 3 for chip financing

Pricing Date	Issuer	Size (\$bn)	UoP	Cost	Offtake	Ratings	Maturity Date
5/15/2026	CRWC	3.1	GPU	SOFR+450	AI Lab / Cohere Inc	Ba2/BB+	11/17/2031
7/24/2026	QTS	3.3	Data Center	SOFR+225	QTS	Baa3/BBB	7/25/2033
8/7/2026	CRWC	2.6	GPU	SOFR+550	Jane Street / Hudson River Trading	Ba2/BB+	9/30/2031
8/10/2026	LAMBCM	0.9	GPU	SOFR+300	NVIDIA	Baa1	12/31/2030

Source: Bloomberg, Morgan Stanley Research

Securitized Credit: A Delayed Reaction?

Supply in the securitized credit markets of data center assets remained steady in July, with a further \$2.2bn of deals brought to market. Issuance is pacing ahead of 2025 (at \$18bn YTD, levels are about 120% of supply this year vs YTD 2025). Notably, spread movement in ABS was relatively muted when compared to corporate credit markets. We think this comes down to a few factors: 1) securitized credit markets have already reacted once to the supply technical; 2) secondary markets are far less liquid, allowing for less price discovery; 3) while still pacing ahead of 2025, volumes are comparatively less strong in sec credit than in corporates; and 4) assets in securitized credit are stabilized, and so less subjected to noise around construction outcomes or policy changes, which we think contributed to some of the move wider in HY. **We expect that once new issue resumes in the fall, and if secondary trading picks up, spreads in securitized credit will soften from here.**

See more here: [ABS Market Insights: Data Center ABS: Finding Value Amidst A Credit Market Sell-off \(6 Aug 2026\)](#)

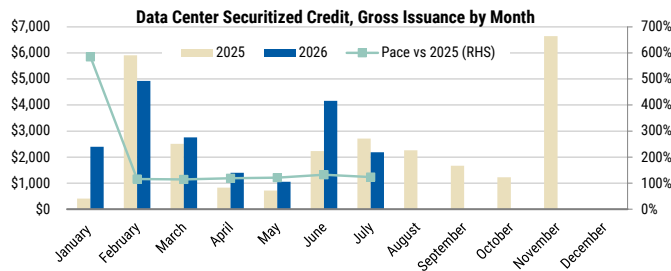
A letter from the SEC and law firm Latham & Watkins confirming "that data center securitizations fall outside the statutory definition of an "asset-backed security" under Section 3(a)(79) of the Securities Exchange Act of 1934, commonly referred to as "Exchange Act ABS." Importantly it is only ABS (because of the inclusion of the asset into and lease-backed nature of the SPV) that fall outside of the regulation, not CMBS (because of the placement of mortgages into the SPV rather than the asset itself). Specifically, the SPV backing ABS deals does not qualify as "self-liquidating", unlike a mortgage, which is part of the definition in ABS. (See Latham's full letter [here](#)).

Most prevalently, falling outside of the scope of the regulations means that sponsors are not beholden to the 5% risk retention rules. Practically speaking, data centers are financed through the ABS market at a loan-to-value rate of often less than 70% but generally no more than 80%. As Latham notes, the issuer is therefore retaining on average 30% risk retention through their own ownership of the equity. If they are not required to hold a minimum 5% risk retention, in theory the sponsor could now sell their entire equity ownership to another market participant.

Other components of the regulation combine to make it marginally more favorable to issue into ABS over CMBS, including limiting some of the other disclosures. To date, it is our understanding that ABS issuers have been complying with the regulation; going forward some of those practices may be relaxed. On the margin we think this makes it more preferable to issue into ABS over CMBS from an issuer perspective, and perhaps less favorable from an investor perspective (given the theoretically relaxed risk retention). However, we think there are greater pressures at work for both issuer and investor selection and that this regulation is down the priority list for decision making.

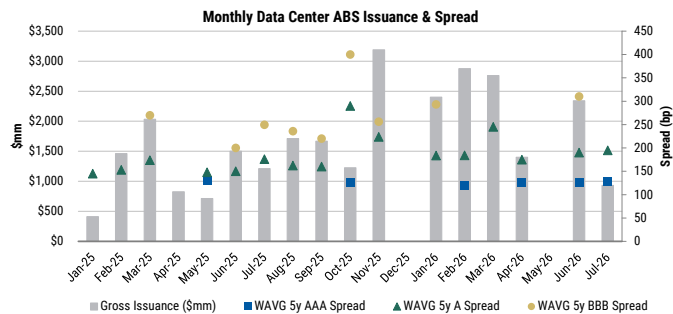
Data Center Securitized Credit Issuance

Exhibit 63: Data center issuance has been pacing about 20% above 2025 so far this year



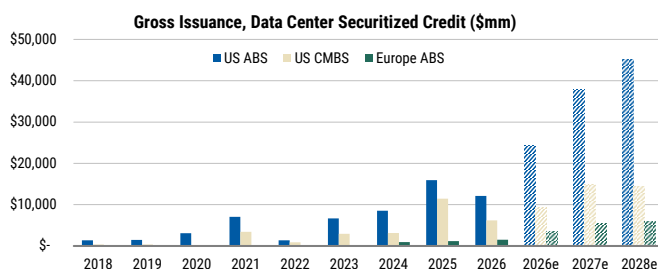
Source: CreditFlow, Intex, Morgan Stanley Research

Exhibit 65: The cadence of issuance has been fairly steady in ABS this year, and spread move fairly tame



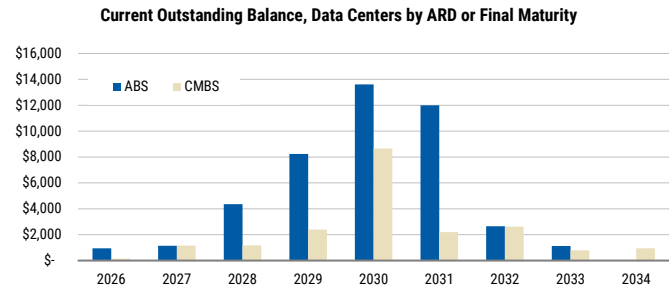
Source: CreditFlow, Intex, Bloomberg, Morgan Stanley Research

Exhibit 67: Gross issuance & forecast, data center securitized credit



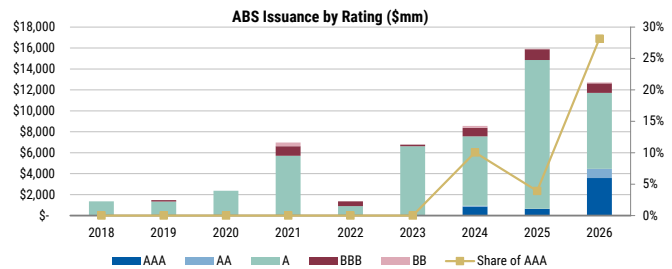
Source: CreditFlow, Trepp, Intex, Morgan Stanley Research

Exhibit 64: While incremental issuance continues to add to the 2030-2031 maturity wall that has been building for some time



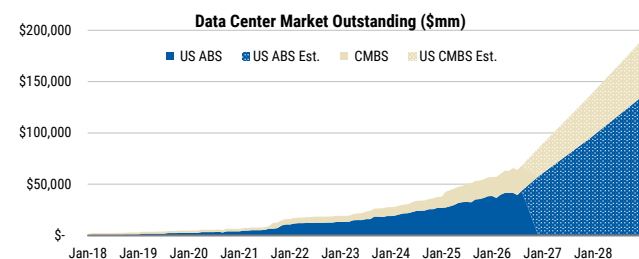
Source: CreditFlow, Intex, Bloomberg, Morgan Stanley Research

Exhibit 66: Increasingly issuers are bringing AAA-rated paper to the market in an effort to crowd in more demand



Source: CreditFlow, Intex, Bloomberg, Morgan Stanley Research

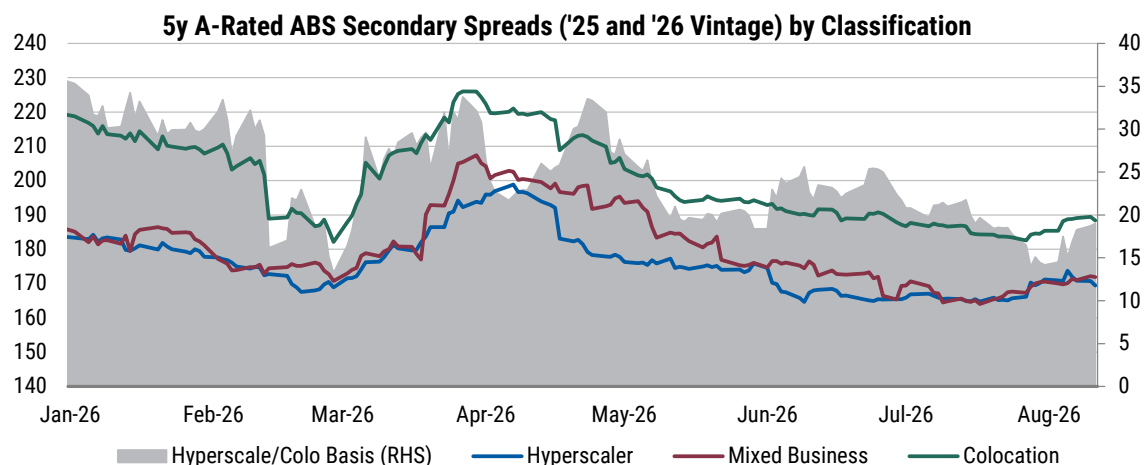
Exhibit 68: Outstanding balance & forecast, data center securitized credit



Source: CreditFlow, Trepp, Intex, Morgan Stanley Research

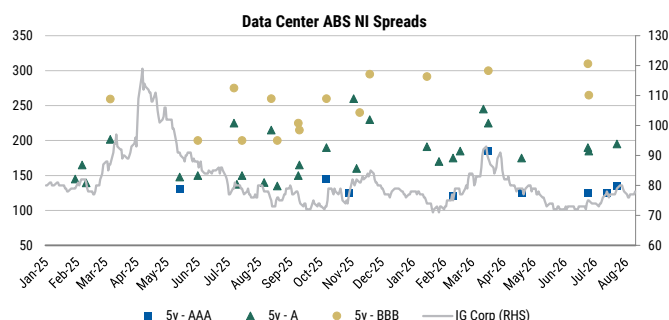
Data Center ABS Spread Performance

Exhibit 69: The basis between colocation & hyperscaler spreads moves largely in tandem with the direction of spreads, tightening throughout the course of the year.



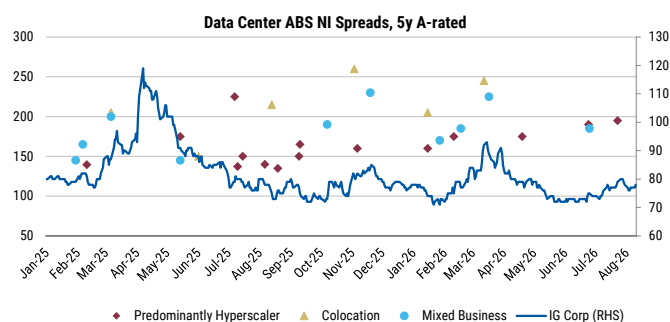
Source: Bloomberg, Morgan Stanley Research

Exhibit 70: New issue data center ABS Spreads (5y WAL)



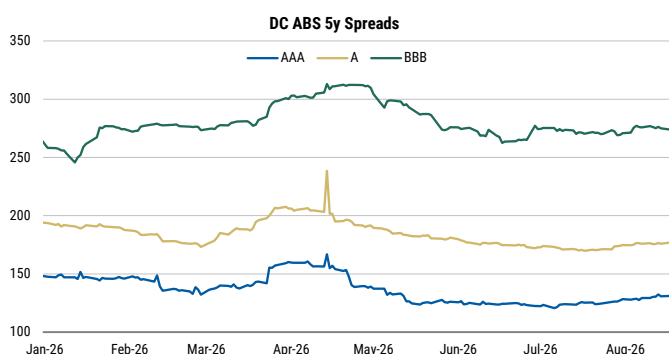
Source: CreditFlow, Morgan Stanley Research

Exhibit 71: New issue data center spreads, 5y WAL, A-rating, by classification



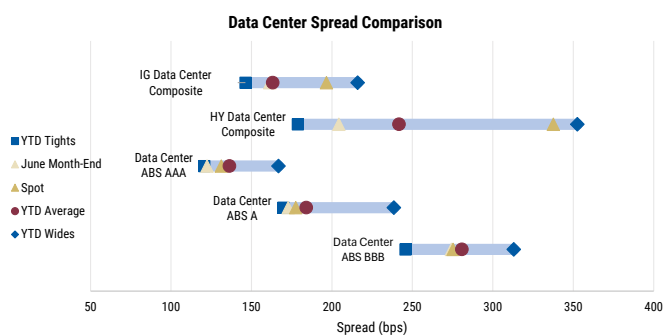
Source: CreditFlow, Morgan Stanley Research

Exhibit 72: Secondary spreads, data center ABS, 5y WAL, by rating



Source: Bloomberg, Morgan Stanley Research. Includes only 2025 and 2026 vintage.

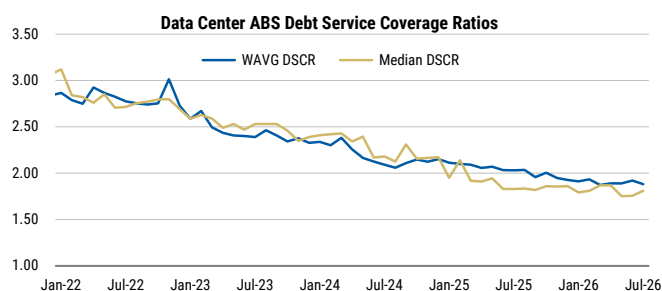
Exhibit 73: Data center ABS spreads vs secured data center corporate credit spreads



Source: Bloomberg, Morgan Stanley Research

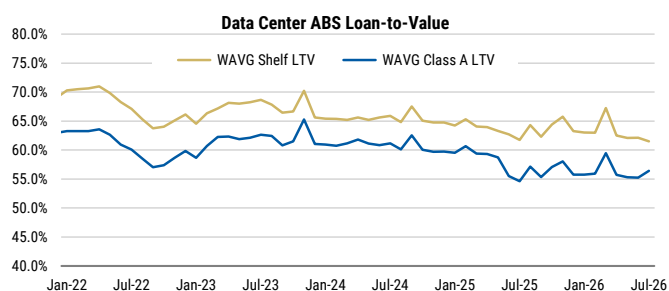
Data Center ABS Performance Metrics

Exhibit 74: Data center ABS Debt Service Coverage Ratios



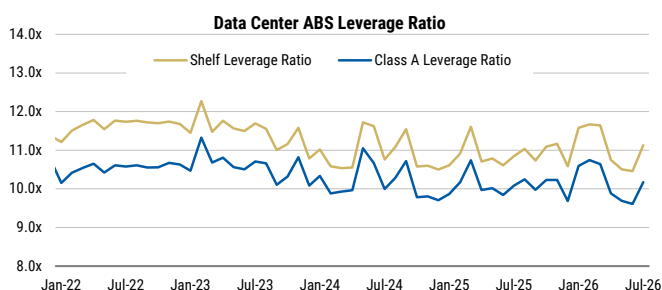
Source: Intex, Morgan Stanley Research

Exhibit 75: Data Center ABS Loan-to-Value Ratios



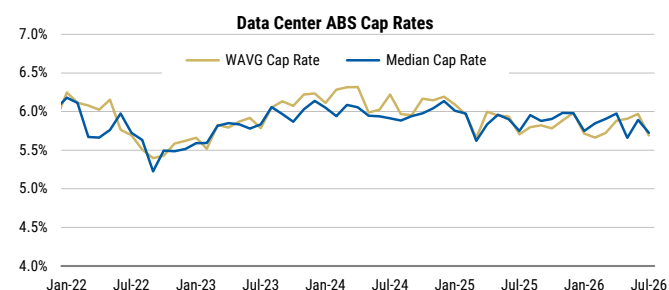
Source: Intex, Morgan Stanley Research

Exhibit 76: Data Center ABS Leverage Ratios



Source: Intex, Morgan Stanley Research

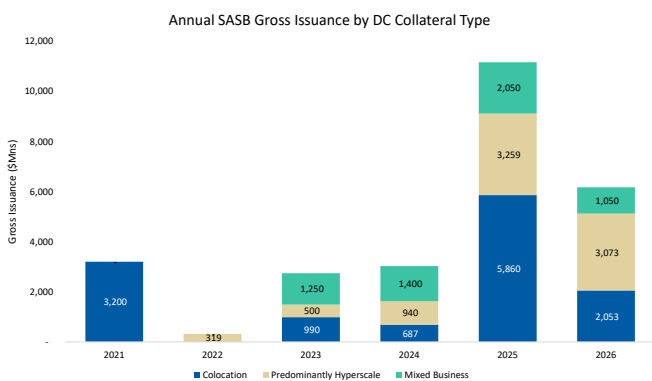
Exhibit 77: Data Center ABS Cap Rates



Source: Intex, Morgan Stanley Research

Data Center CMBS New Issue Credit Metrics & Pricing

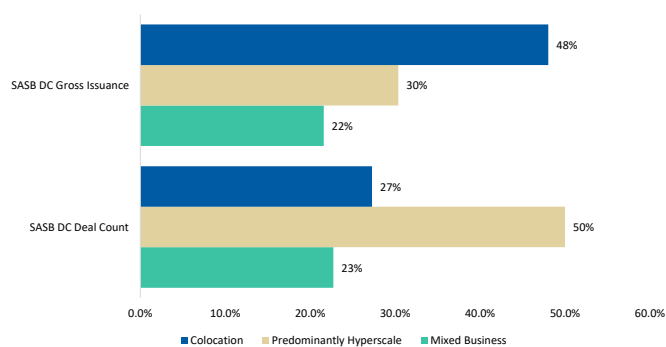
Exhibit 78: SASB data center issuance has reached \$6.2bn YTD across 5 transactions; we expect DC issuance to rebound after August lull



Source: Trepp, Morgan Stanley Research

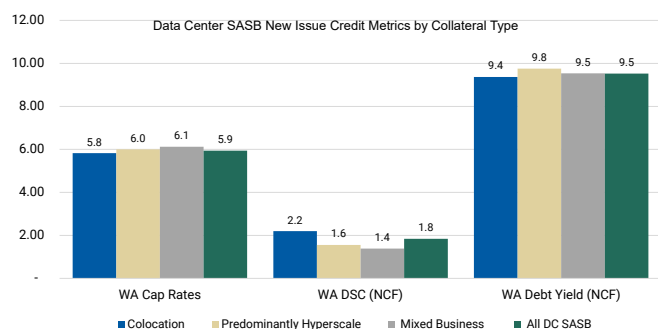
Exhibit 79: Predominantly hyperscaler-backed deals accounting for the majority of 2026 volume

Composition of SASB Data Center Issuance by Collateral Type



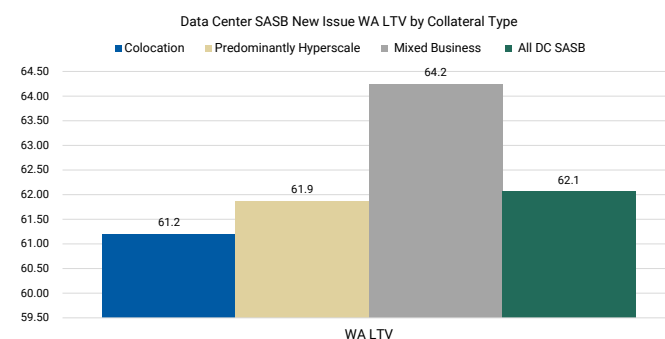
Source: Trepp, Morgan Stanley Research

Exhibit 80: Cap rates for 2026 DC SASBs have ranged from 6.0% to 6.6%



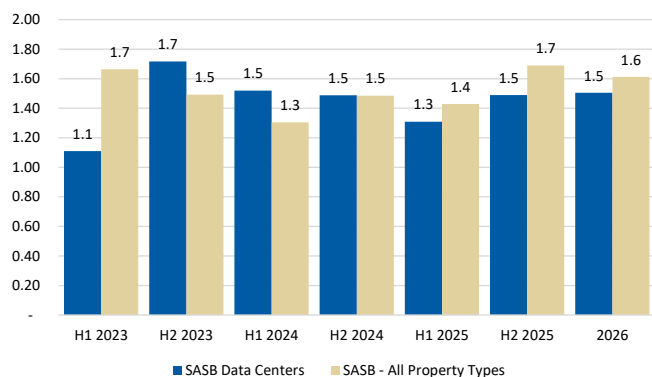
Source: Trepp, Morgan Stanley Research

Exhibit 81: Mixed business reports the highest leverage among DC collateral types



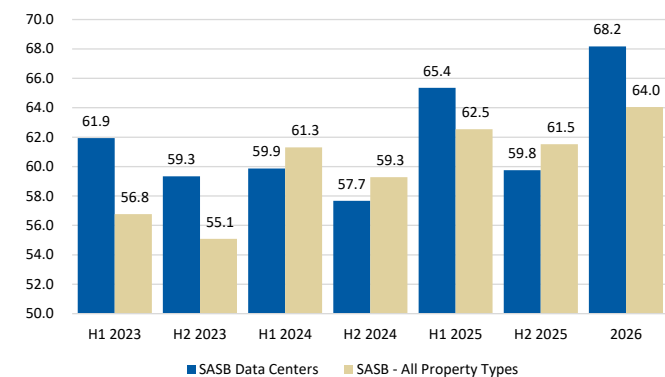
Source: Trepp, Morgan Stanley Research

Exhibit 82: 2026 DC SASBs report a WA debt service coverage ratios of ~1.5x



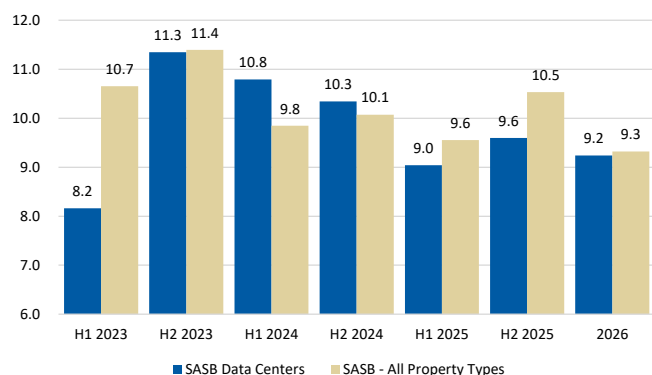
Source: Trepp, Morgan Stanley Research

Exhibit 83: Average LTV for 2026 data center SASBs increased to 68.2%, above the broader SASB average



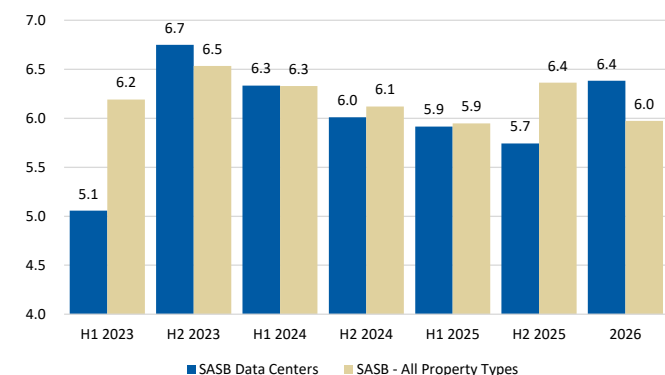
Source: Trepp, Morgan Stanley Research

Exhibit 84: Debt yields declined across 2026 SASB issuance, including data center transactions, relative to H2 2025 levels

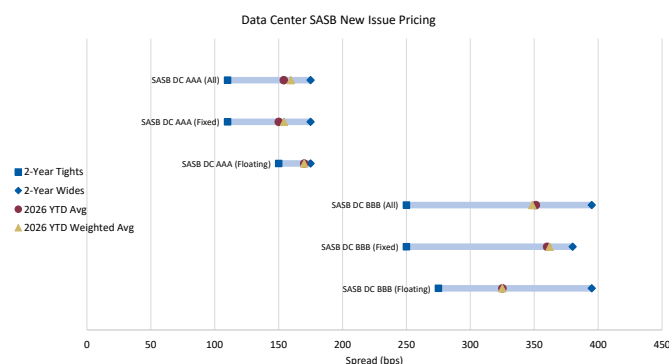


Source: Trepp, Morgan Stanley Research

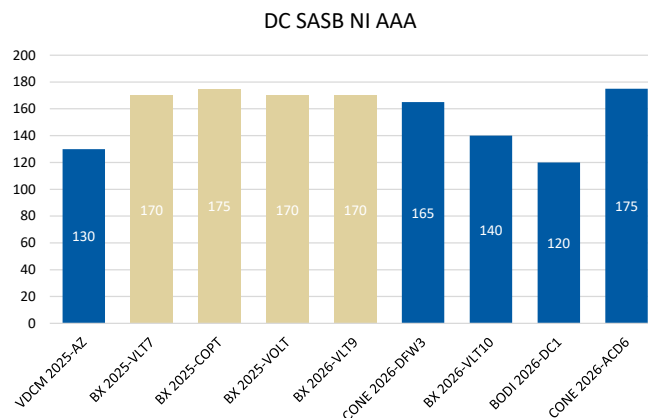
Exhibit 85: Cap rates on 2026 data center SASBs have widened to ~6.4% vs 2025 vintages



Source: Trepp, Morgan Stanley Research

Exhibit 86: New issue SASB DC Spreads - YTD average vs 2yr tights and wides

Source: Green Street Advisors; Morgan Stanley Research

Exhibit 87: DC SASB New Issue AAA Spreads

Source: Trepp, Morgan Stanley Research; blue bars denote fixed rate SASBs, yellow bars denote floating rate SASBs

Data Center ABS and CMBS Recent Pricings

Exhibit 88: New issue data center ABS

ABS New Issue Market													
Deal	Tranche	CUSIP	Pricing Date	ARD	WAL	Rating	Original Size	Current Size	Year	Spread	Coupon	Current YTM	LTV at Issuance
CMDC 2026-2	CMDC 2026-2 A2I	20469BAJ6	7/24/2026	7/25/2031	5.0	AAA	295	295	2026	135	5.72%	5.69%	38.90%
CMDC 2026-2	CMDC 2026-2 A2II	20469BAK3	7/24/2026	7/25/2031	5.0	AA-	68	68	2026	170	6.06%	6.00%	48.10%
CMDC 2026-2	CMDC 2026-2 A2III	20469BAL1	7/24/2026	7/25/2031	5.0	A	50	50	2026	195	6.30%	6.24%	54.40%
CLDCP 2026-1	CLDCP 2026-1 A2I	54143VAA4	7/14/2026	7/15/2031	5.0	AAA	520	520	2026	125	5.12%	5.58%	38.80%

Source: CreditFlow, Intex, Bloomberg, Morgan Stanley Research

Exhibit 89: New issue data center CMBS

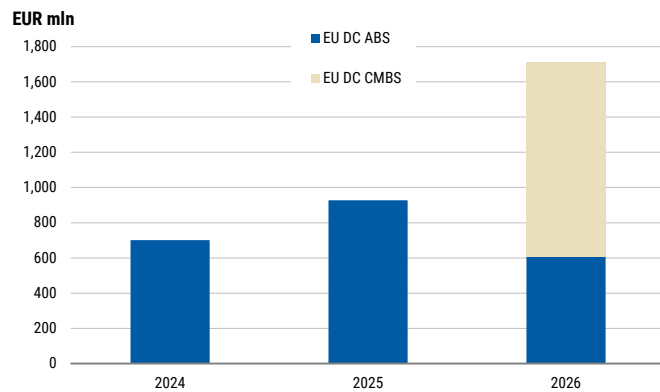
SASB New Issue Market											
Deal Name	Tranche	Cusip	Pricing Date	Class	Tranche Size (\$mn)	Spread	Coupon	Rate Type	Current Mid YTM	Min Original Rating	
CONE 2026-ACD6	CONE 2026-ACD6 A Mtge	12637SAA2	7/1/2026	A	\$782	175	6.06%	Fixed	6.08%	AAA	
CONE 2026-ACD6	CONE 2026-ACD6 B Mtge	12637SAC8	7/1/2026	B	\$116	220	6.52%	Fixed	6.74%	AA	
CONE 2026-ACD6	CONE 2026-ACD6 C Mtge	12637SAE4	7/1/2026	C	\$109	265	6.97%	Fixed	7.25%	A-	
CONE 2026-ACD6	CONE 2026-ACD6 D Mtge	12637SAG9	7/1/2026	D	\$181	380	8.11%	Fixed	8.43%	BBB	
CONE 2026-ACD6	CONE 2026-ACD6 HRR Mtge	12637SAJ3	7/1/2026	HRR	\$63		11.53%	Fixed		BBB-	

Source: Bloomberg, Trepp, Morgan Stanley Research

European DC ABS & CMBS

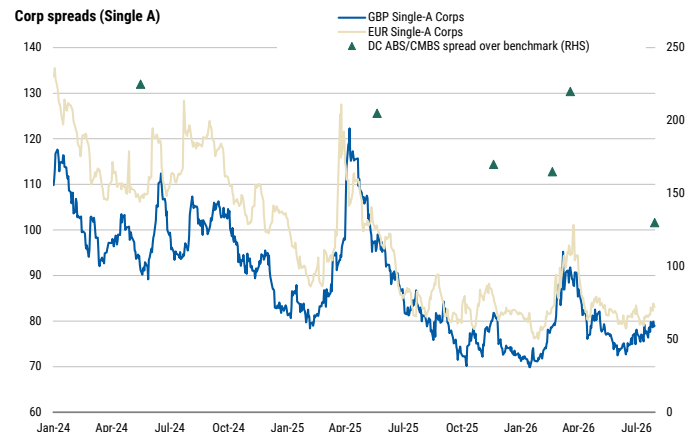
The European data centre securitization space is small for now, comprising a little over €3bln in current outstanding size from three ABS transactions and two CMBS deals. The absolute numbers are small but the momentum seems to be picking up - three new data centre operators have issued a transaction this year. Issuance in 2026 aggregates €1.7bln so far and has included the sector's first CMBS structures as well.

Exhibit 90: European DC securitized issuance is modest but growing



Source: Bloomberg, Morgan Stanley Research. Note: Our CMBS numbers include the Equinix xScale deal which has been issued as a secured bond.

Exhibit 91: EU DC ABS/CMBS new issue spreads vs IG corporate spreads

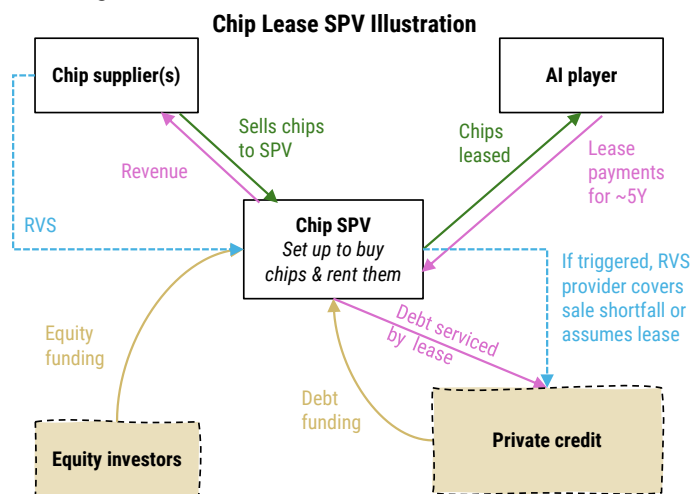


Source: Bloomberg, Markit, Morgan Stanley Research

Single Name Coverage: What's Topical Of Late in IG TMT

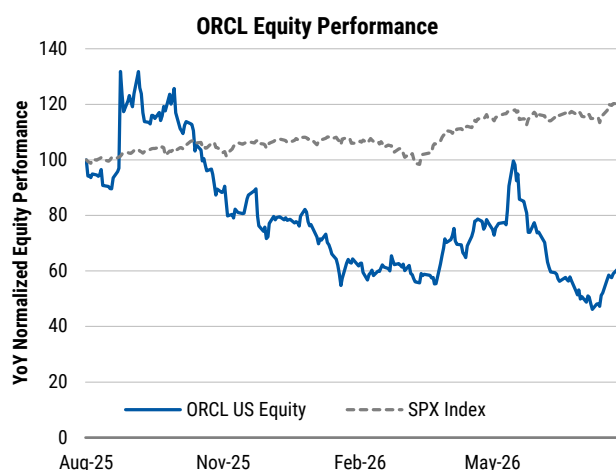
Our single-name views and investor inbound remain focused on AI financing, particularly capacity, structures, and support. ORCL and AVGO remain front and center ahead of their respective September earnings prints. Recent chip/compute financing developments reinforce the growing role of highly rated chip vendors in enabling AI financing, while reigniting debates around circularity.

Exhibit 92: Chips: Residual value support from an IG player can enhance scale and pricing for a chip lease-backed SPV debt financing



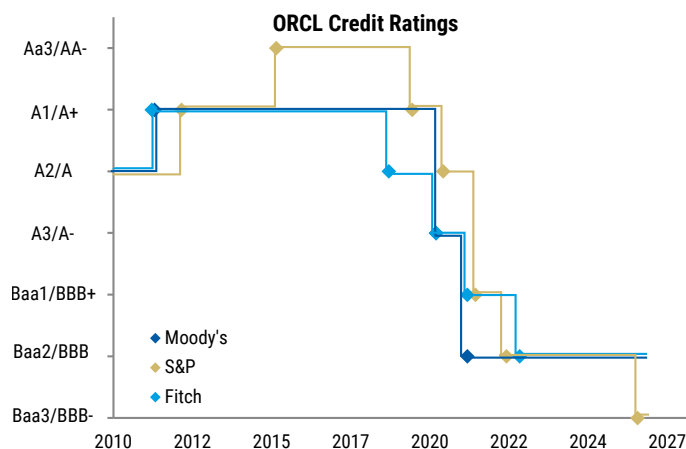
Source: Company filings, Bloomberg, Morgan Stanley Research

Exhibit 93: ORCL: We're watching not only debt costs but also equity performance, as we await clarity on ATM utilization in the FY1Q27 earnings print and assess ongoing viability of the equity lever



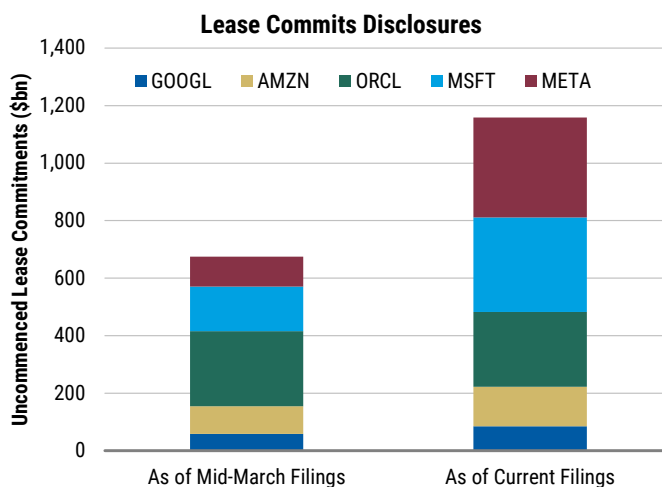
Source: Bloomberg, Morgan Stanley Research

Exhibit 94: ORCL: Lower starting ratings and a subsequent S&P downgrade underscore a more differentiated and constrained path vs. HQ hyperscalers



Source: Moody's, S&P, Fitch, Bloomberg, Morgan Stanley Research. Read more: [ORCL Credit: Up the Spending Scale, Down the Rating Scale \(9 Jul 2026\)](#).

Exhibit 95: ORCL: Lease commitments now exceed \$1tn across major IG hyperscalers, with ORCL no longer the largest contributor



Source: Company filings, Morgan Stanley Research. Read more: [ORCL Credit: \\$260bn Off Balance Sheet, But On Our Mind \(22 Jun 2026\)](#).

TRADE	ENTRY LEVEL	ENTRY DATE	CURRENT LEVEL	RATIONALE	RISKS
Buy ORCLCP 5Y CDS	+57bp	9/28/25	+217bp	• Credit profile deterioration, driven by cash funding needs and increasing lease liabilities related to new data centers • Lender hedging may be a key driver of CDS too • For relative value, we've considered other large structure situations where there have been ratings risk	• Constructive updates on RPO/growth and financing clarity • TL As may not ultimately serve a significant funding role • CDS technicals: Basis selling, construction loan hedge unwinds, fast money outright shorts profit taking
Sell ORCL 5.2 9/26/35	+167bp	1/23/26	+203bp	• Credit profile deterioration, driven by cash funding needs and increasing lease liabilities related to new data centers • For relative value, we've considered large media/cable structure comps	• Additional visibility on long-term monetization and delevering beyond FY28-end • Issuance cadence may be different than our assumptions
Sell ORCL 5.95 9/26/55	+208bp		+258bp		

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(as of July 31, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1529	42%	442	47%	29%	750	43%
Equal-weight/Hold	1582	43%	396	42%	25%	773	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	560	15%	97	10%	17%	217	12%
Total	3,674		936			1741	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Equal-weight (E or Equal) - The stock's total return is expected to be in line with the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Not-Rated (NR) - Currently the analyst does not have adequate conviction about the stock's total return relative to the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U or Under) - The stock's total return is expected to be below the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

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Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

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